SMRTTECH 4SC3

Smart Cities and Communities Lab

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Week 3:
Digital Twin in Smart Cities I

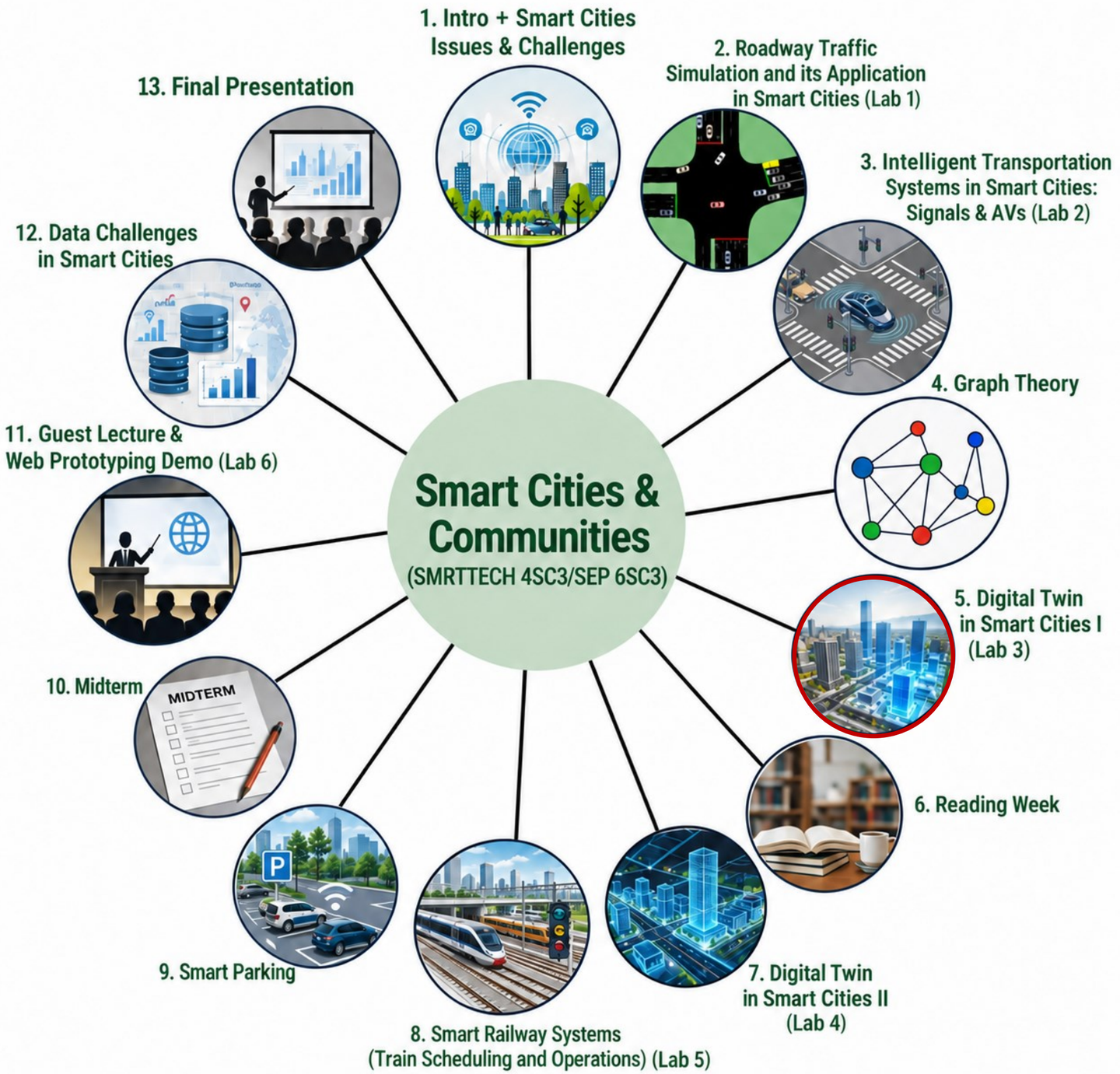
Fall 2026

McMaster University
Engineering
W Booth School of Engineering Practice and Technology



Agenda

- ❑ Fundamentals of 3D Modelling
- ❑ Blender (Quick Start)
- ❑ Download Required Materials
- ❑ Milestone 1
- ❑ Deliverable 1
- ❑ Milestone 2
- ❑ Deliverable 2
- ❑ Milestone 3 (optional)



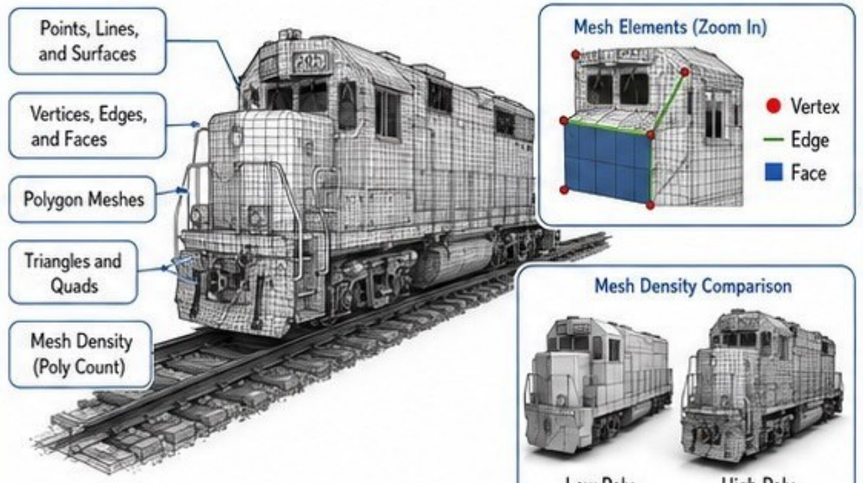
Illustrative Agenda: Fundamentals of 3D Modelling

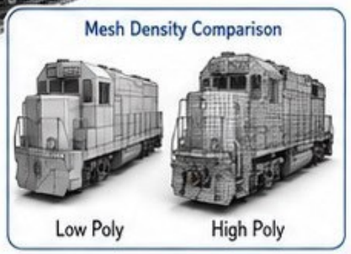
1 Real Object vs Digital Object



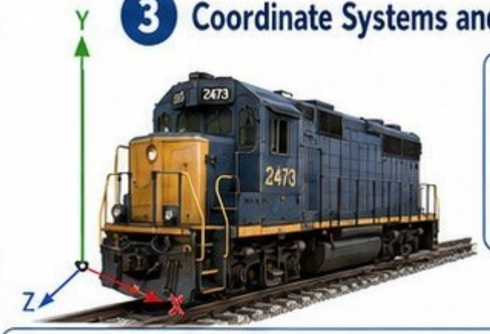
A 3D model is a digital representation of a real object.

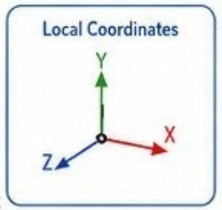
2 Geometry: The Body of a 3D Model

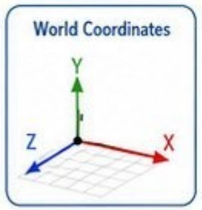


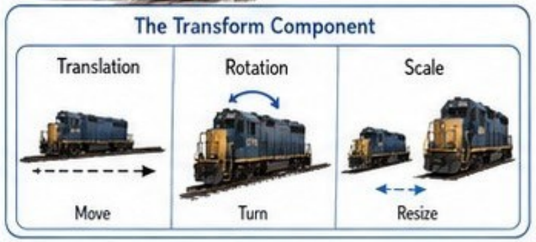


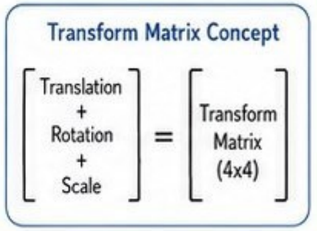
3 Coordinate Systems and Transform

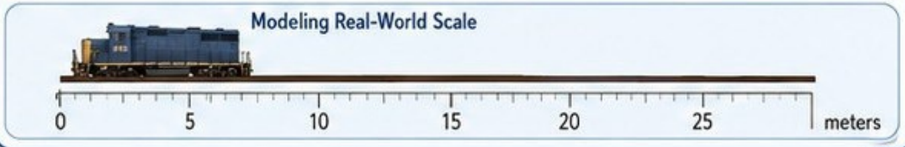




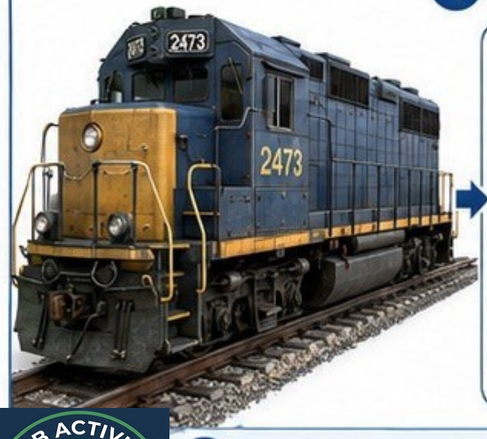




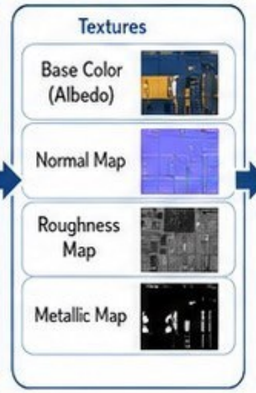


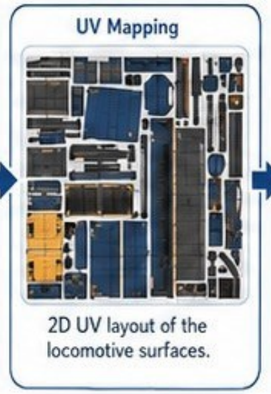


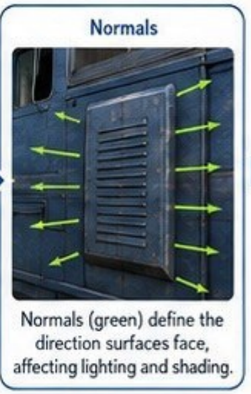
4 Appearance and Surface Definition











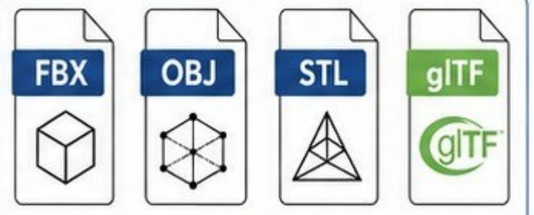
Materials and textures define appearance; UVs and normals control surface mapping and lighting.

5 File Formats



EXPORT

IMPORT



Formats store geometry and sometimes materials, animation, and metadata.

Blender in Digital Twin Development

Blender mainly supports the Model Layer by creating 3D geometry and visual representations of real-world assets.

What is Blender?

- 3D modeling and visualization software
- Create, edit, texture, animate, render

Main Role in Digital Twin

Model Layer

- Builds 3D geometry
- Creates visual representations of real assets

How Blender Helps

- Converts real-world objects into 3D assets
- Creates buildings, roads, rail corridors, stations, and vehicles
- Adds materials, textures, lighting, and realistic details
- Prepares models for Unity or Unreal Engine



Real World Asset



3D Asset in Blender



blender

Game Engine / Digital Twin Platform



unity | UNREAL ENGINE

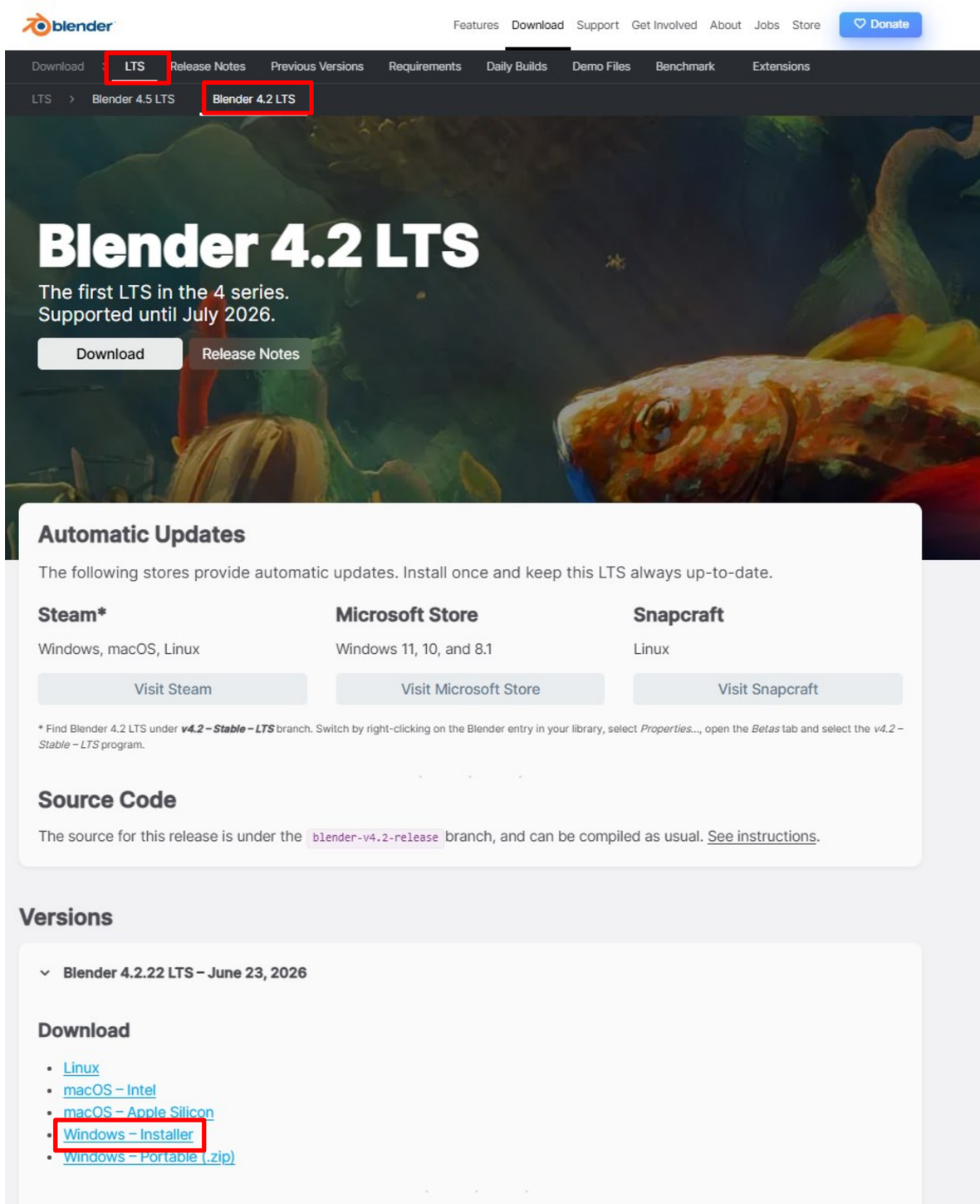
Blender Installation

Task: Install Blender

- ❑ Open The Following Link from Blender official website

  https://www.blender.org/download/lts/

- ❑ Download Blender Version 4.2 LTS

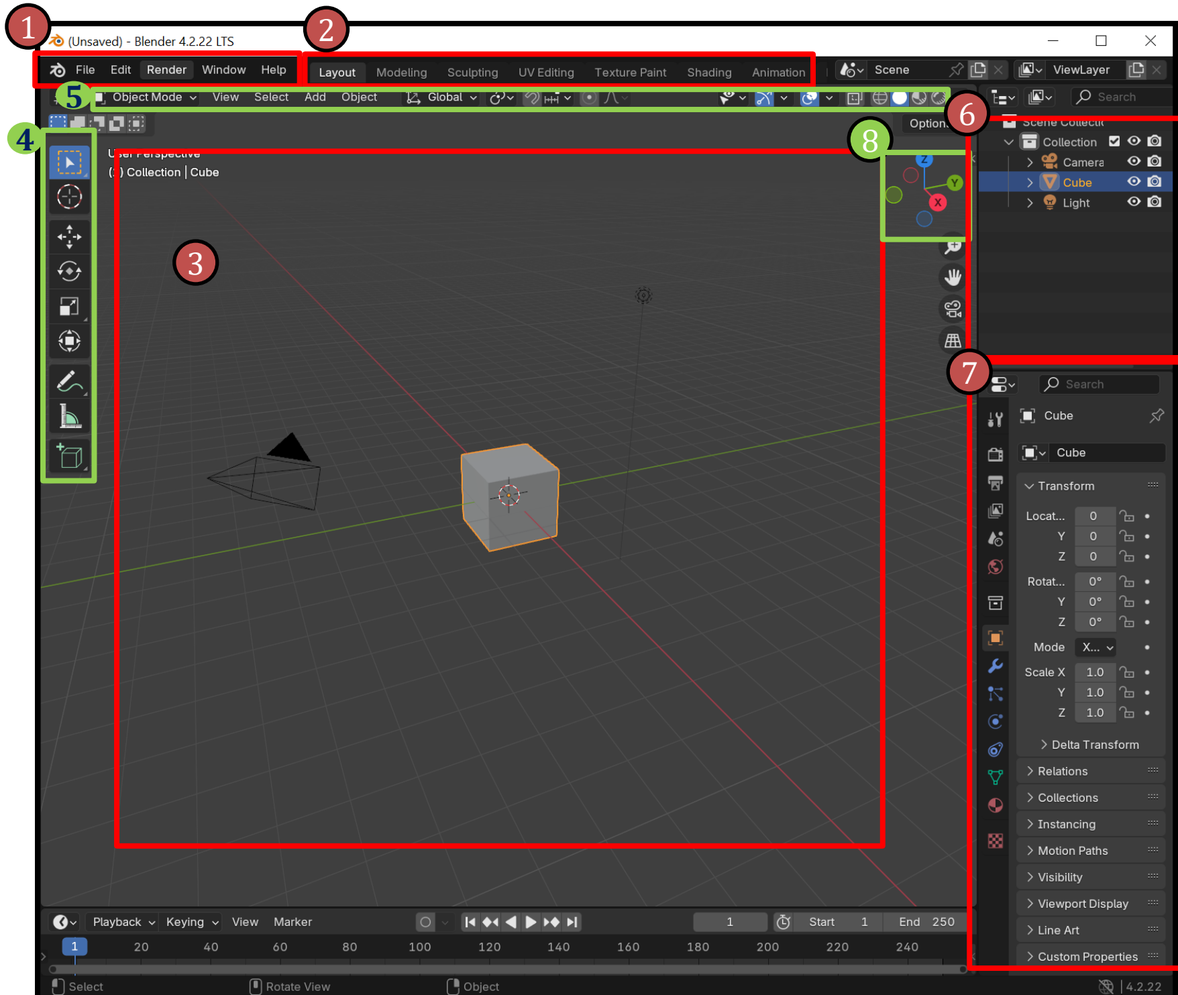


Download Required Materials

- ❑ Download material source file “Railway3D” from GitHub
- ❑ https://github.com/RoadwayVR/SmartCities
- ❑ Extract the zip file
- ❑ Open a New Blender → (no desktop icon, but the executable file is located in C:\Program Files\Blender)

Blender Interface Overview

Blender Area	What It Does
1 Menu Bar	Access file, edit, render, window, and help options.
2 Workspace Tabs	Switch between tasks such as Layout, Modeling, Sculpting, UV Editing, Shading, Animation, and Rendering.
3 3D Viewport	Main workspace where you view, create, and edit 3D objects and scenes.
4 Toolbar	Contains tools for selection, move, rotate, scale, transform, annotate, and other editing actions.
5 Viewport Header	Provides controls for object mode, view, select, add, snapping, orientation, and shading options.
6 Outliner (Scene Collection)	Lists all objects in the scene, such as cameras, lights, meshes, and collections.
7 Properties Editor	Shows and edits settings for the selected object, including transform, materials, modifiers, render, and scene properties.
8 Navigation Gizmo	Helps change view orientation (top, front, side, etc.) inside the 3D viewport.

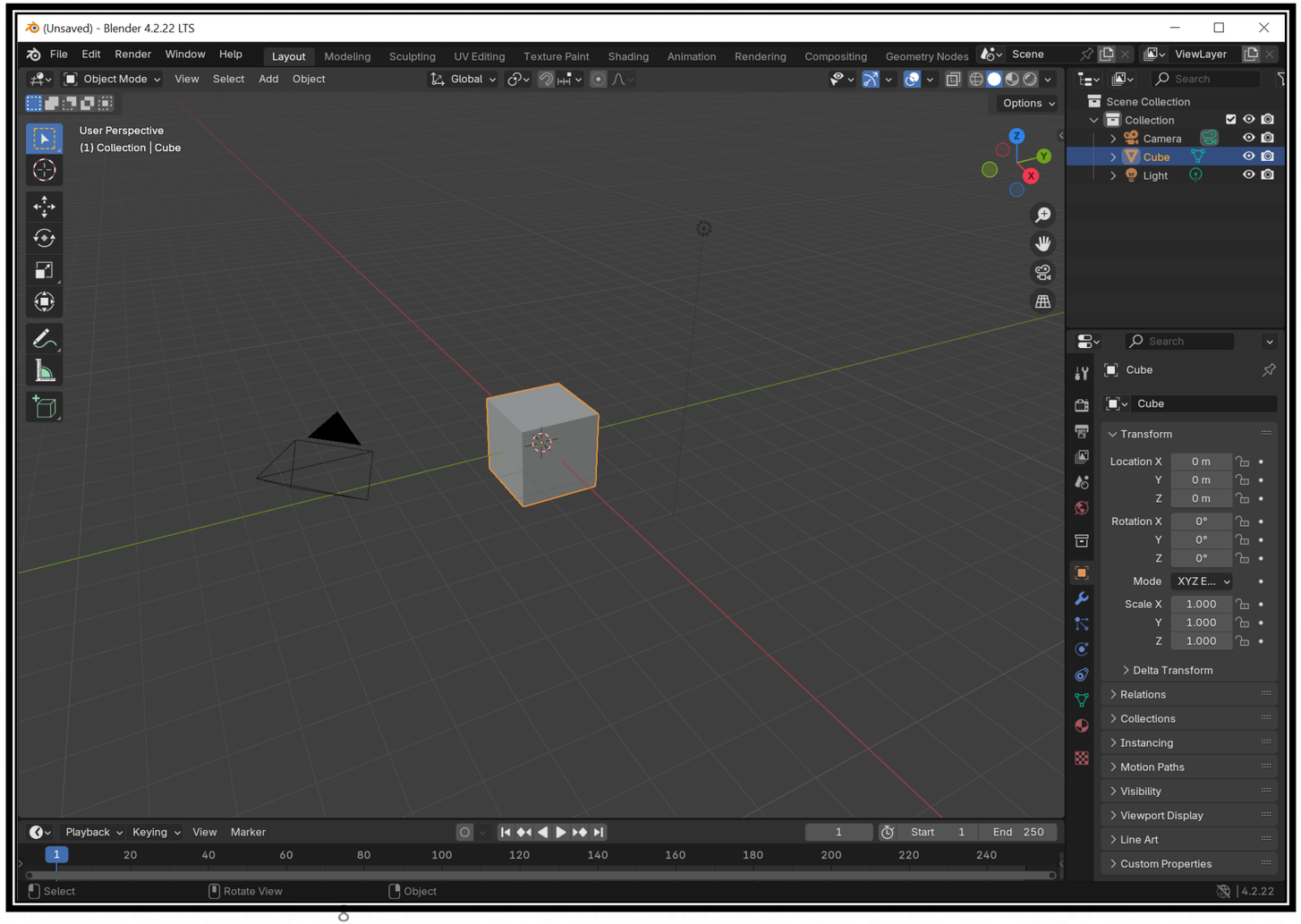


Blender Navigation Control

❑ Practice these 3D Viewport controls for 5 minutes.

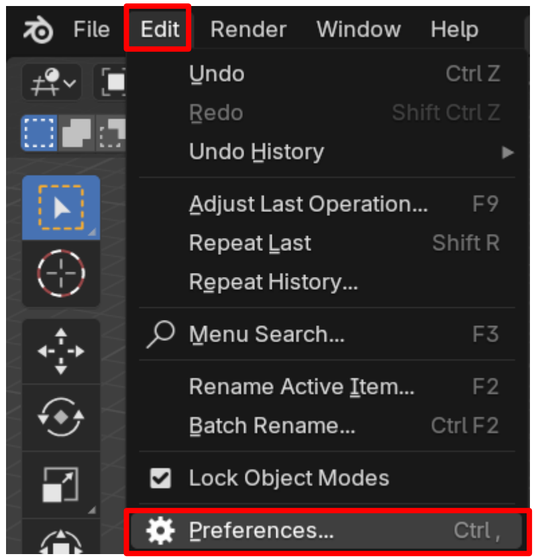
Blender 3D Viewport Navigation Controls

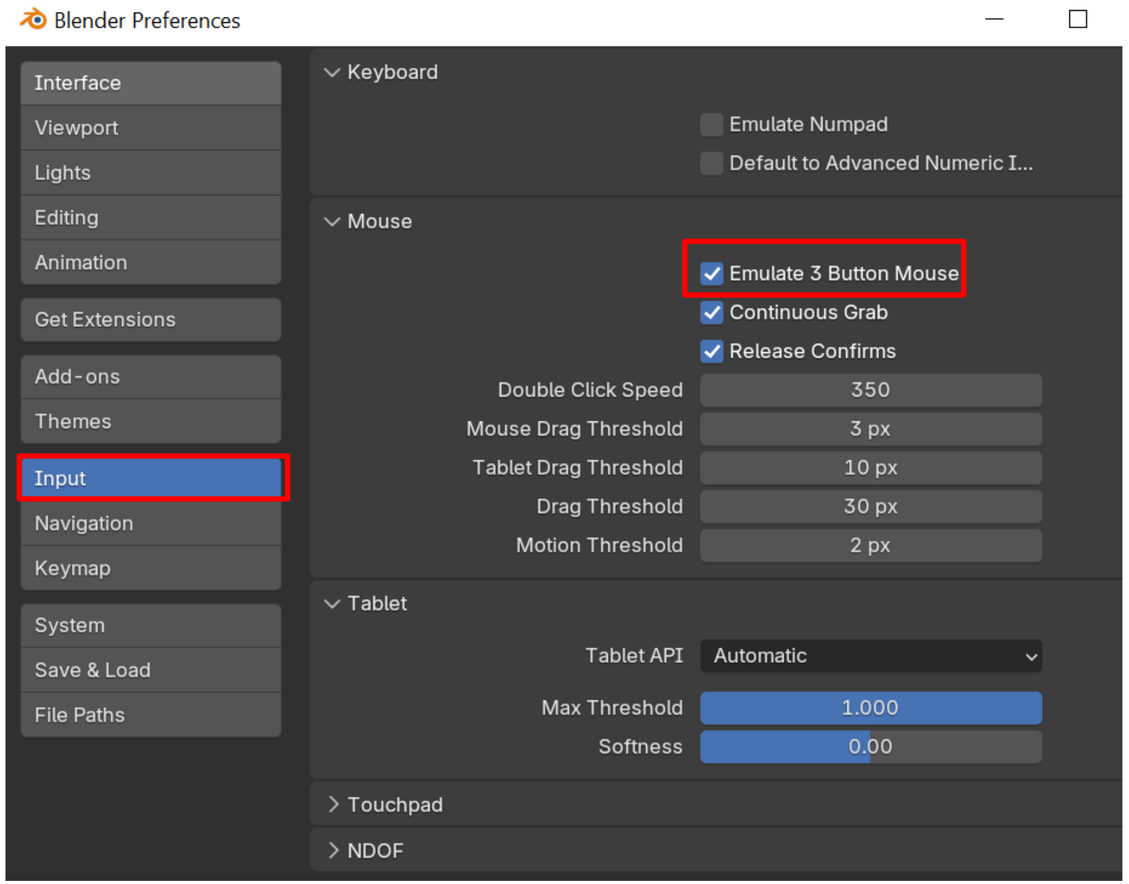
Action	Control
Rotate View	Middle Mouse Button + Drag
Pan View	Shift + Middle Mouse Button + Drag
Zoom In/Out	Scroll Wheel
Look Around / Orbit	Hold Middle Mouse Button
Frame Selected	Numpad .
Front View	Numpad 1
Side View	Numpad 3
Top View	Numpad 7
Camera View	Numpad 0
Walk Navigation	Shift + `



Blender Navigation Control

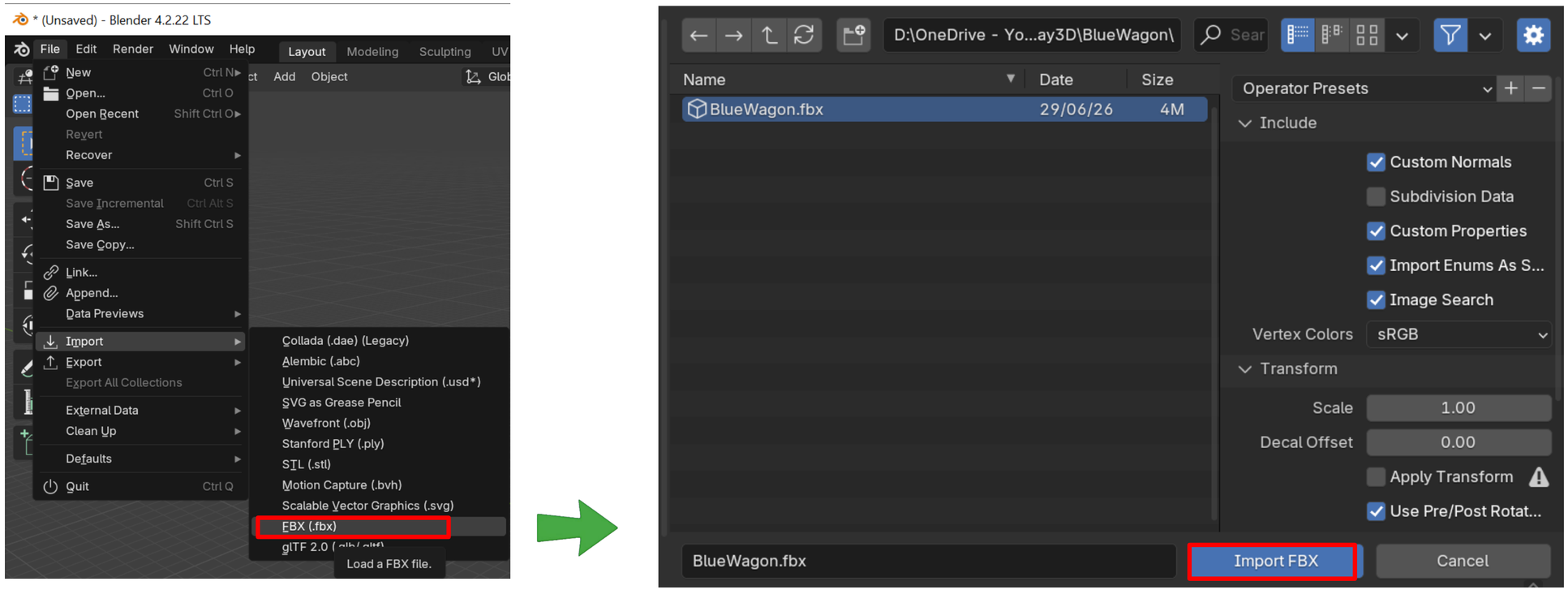
- **Goal:** Alternative button for middle mouse button
- Edit → Preferences → Input → Emulate 3 Button Mouse
- **Task:** Rotate the view by holding “Alt + Left Click Mouse Button”
- **Task:** Pan View by holding “Shift + Alt + Left Click Mouse Button”





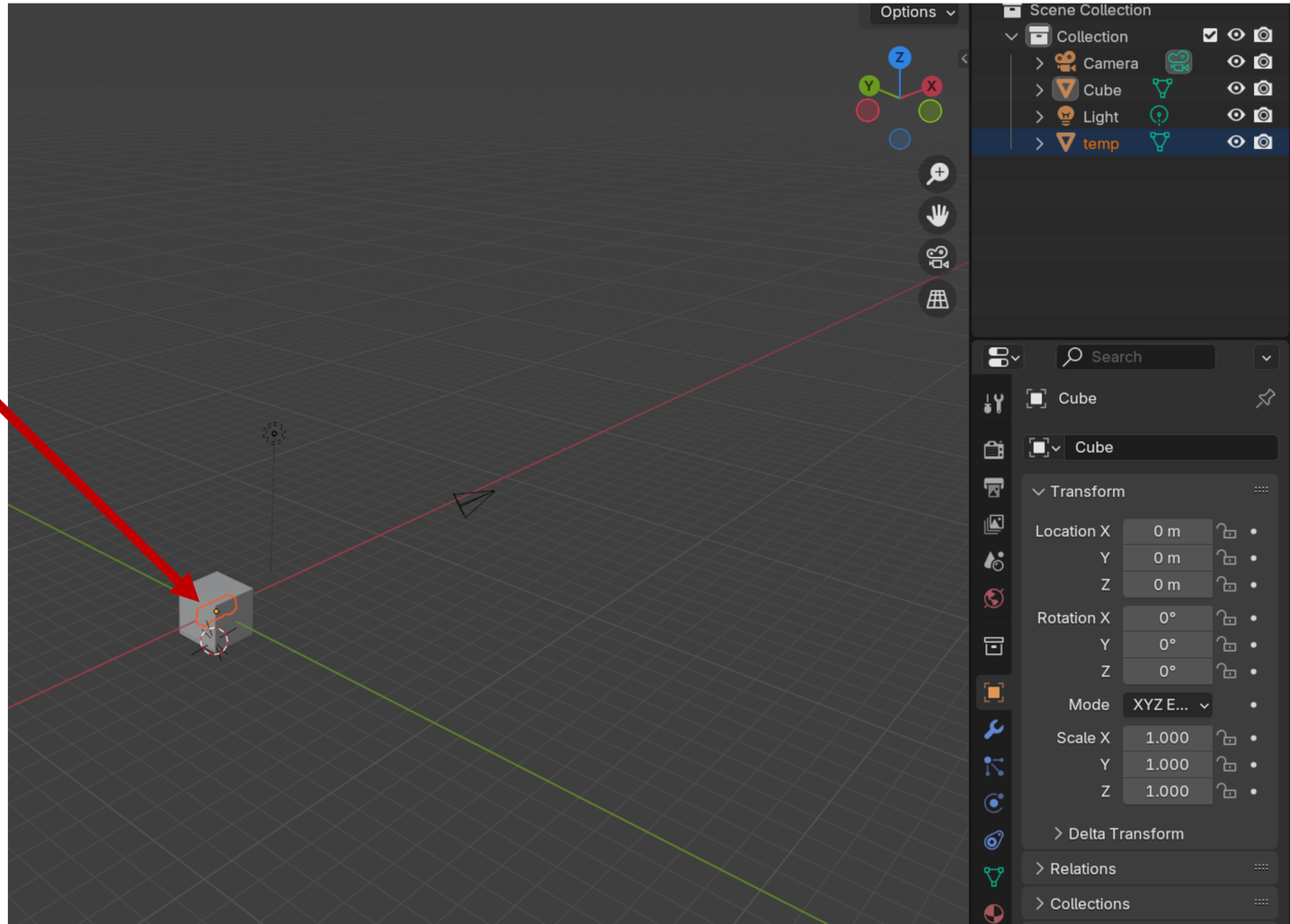
Import Digital Object in Blender

- ❑ File→ Import → FBX
- ❑ Find Folder “BlueWagon” (Lab 3) → Select BlueWagon.fbx → Import it



Import Digital Object in Blender

- ❑ One object called “temp” is added in “Scene Collection”
- ❑ You can visually observe it in the scene
- ❑ The object is inside the Cube!
- ❑ **Task:** hide the Cube (**search it – Use Chatpgt**)



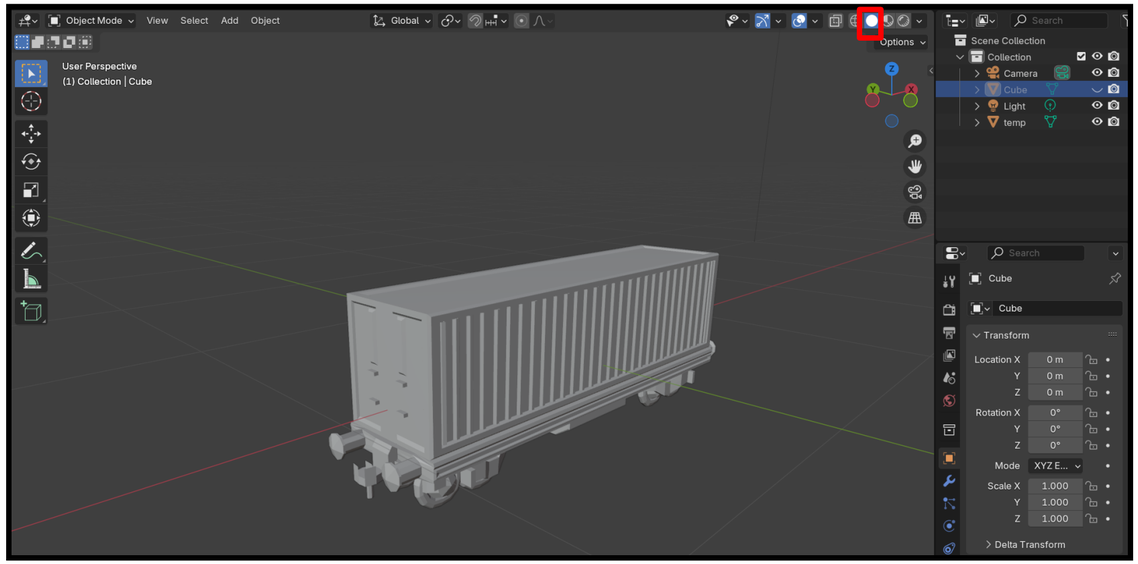
Real Object vs Digital Object

- ❑ Zoom into the imported object and look at the object exactly from the perspective in the image
- ❑ Viewport Header → Select “Viewport Shading: Material Mode Preview”
- ❑ Return to Solid Mode Preview Again

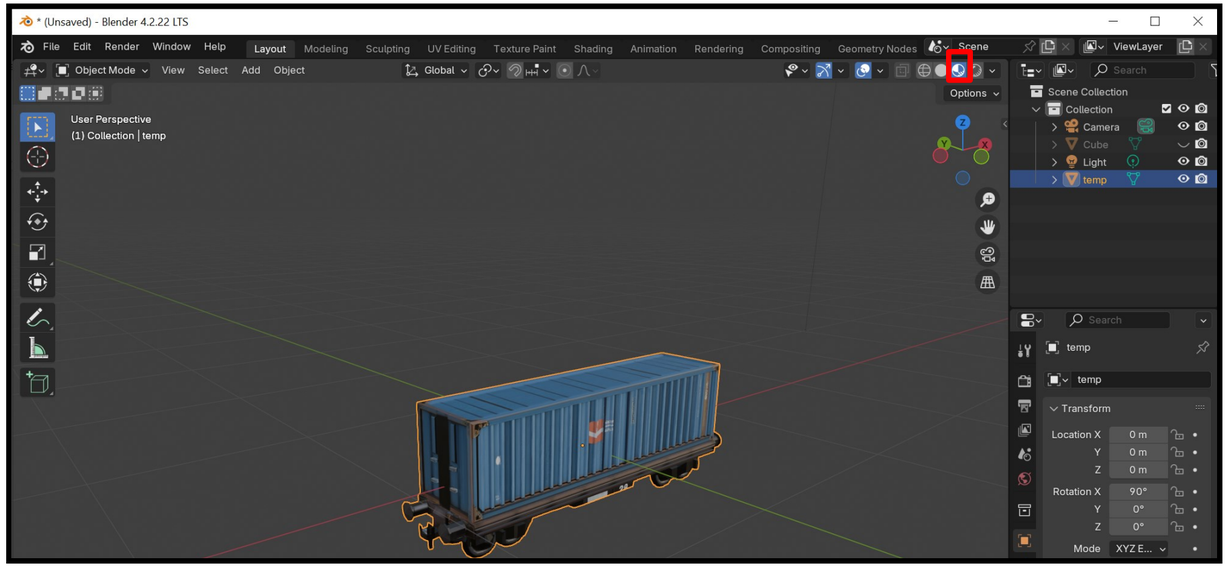
Real Object



Digital Object: Solid Mode Preview

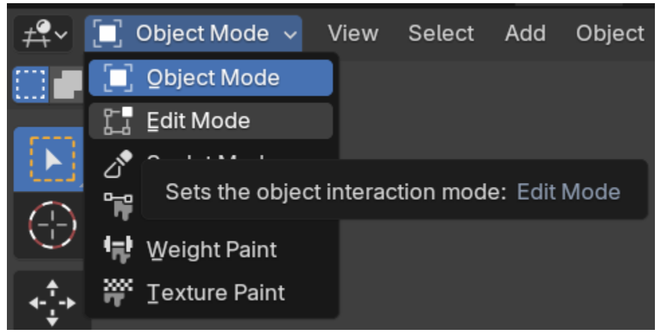


Digital Object: Material Mode Preview

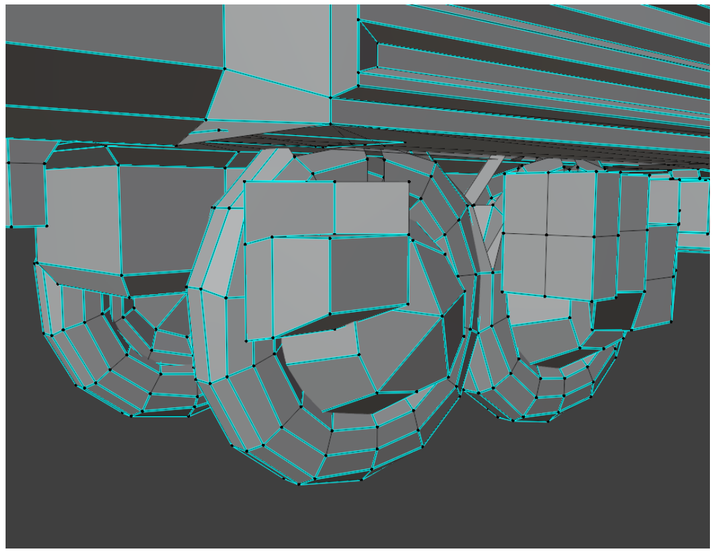


Geometry: The Body of 3D Mesh

- ❑ Switch to “Edit Mode”
- ❑ Zoom into “The Front Left Wheel” exactly like the image
- ❑ **Task:** Identify Points (Vertices), Lines(Edges), Surfaces (Faces)
- ❑ **Polygon Mesh:** Lists of Vertices and How they Connect Into Faces

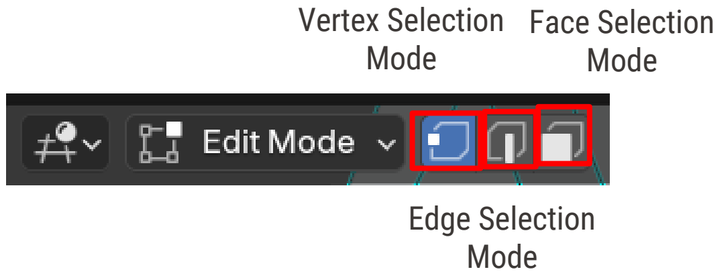


Polygon Mesh including Points (Vertices), Lines(Edges), Surfaces (Faces)



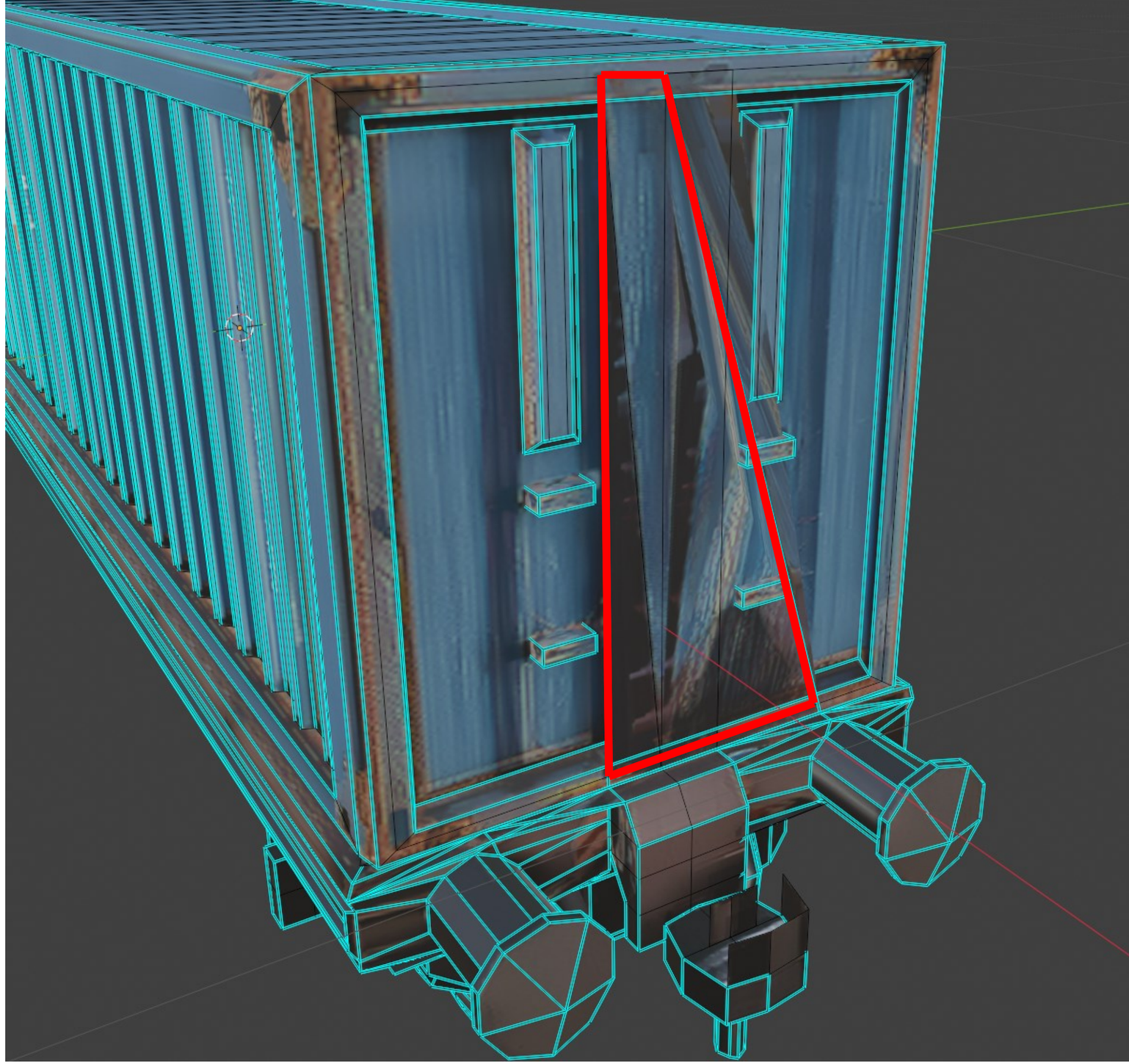
Geometry: Identifying Points (Vertices), Lines (Edges), Surfaces (Faces)

- ❑ Switch to “Vertex Selection Mode” and Select one Vertex in the Wheel, Remove that Vertex by Pressing “Delete” Button → then Undo, “Control Z”
- ❑ Switch to “Edge Selection Mode” and Select one Edge in the Wheel, Remove that Edge by Pressing “Delete” Button → then Undo, “Control Z”
- ❑ Switch to “Face Selection Mode” and Select one Face in the Wheel, Remove that Edge by Pressing “Delete” Button → then Undo, “Control Z”



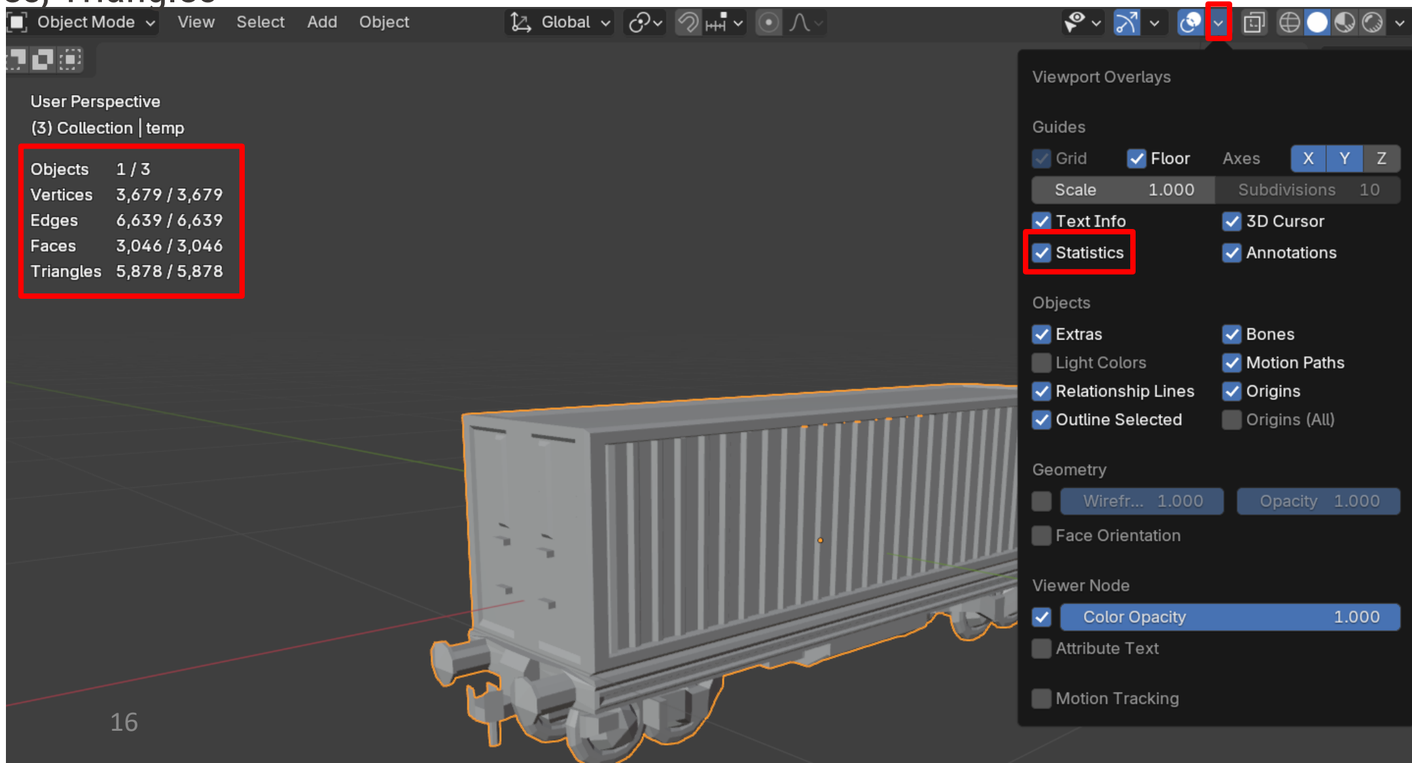
Checkpoint

- ❑ Sometimes, we have redundant faces in our digital object (See Red Lines).
- ❑ Remove redundant faces
- ❑ Are these faces Triangle or Quad Polygon Mesh?
- ❑ Raise Your Hand When You Finished



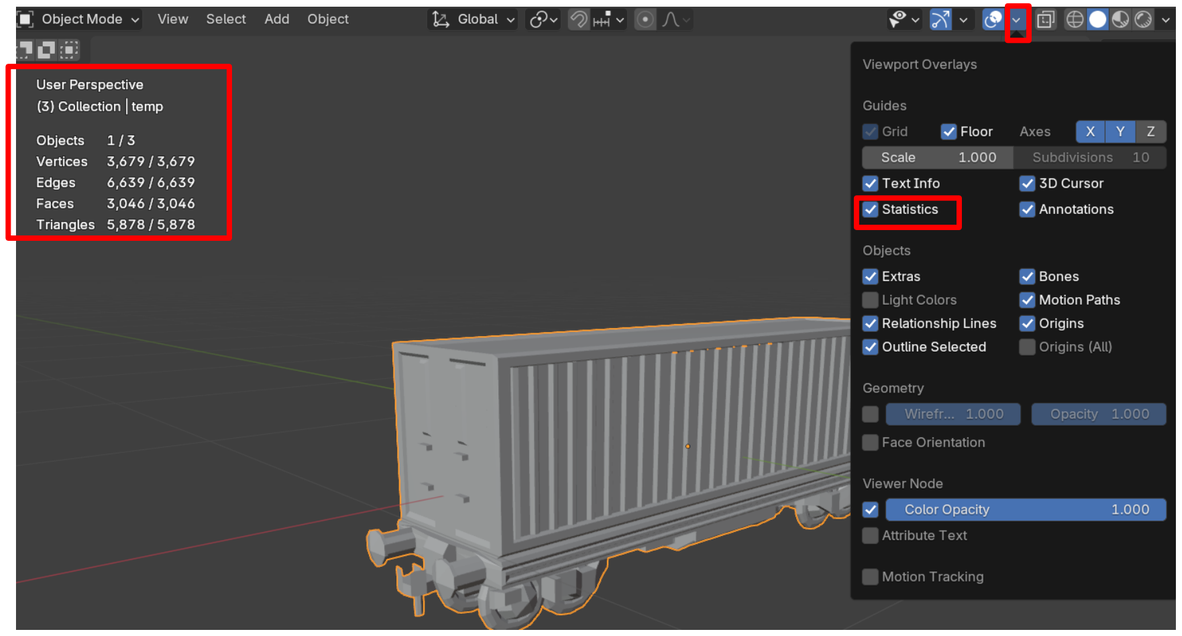
Mesh Density

- ❑ **Goal:** Count the Number of Vertices, Edges, faces, Triangles
- ❑ Select object “Wagon”
- ❑ Select “Overlays” → Check “Statistics”
- ❑ Read the Statistics about Number of Vertices, Edges, Faces, Triangles

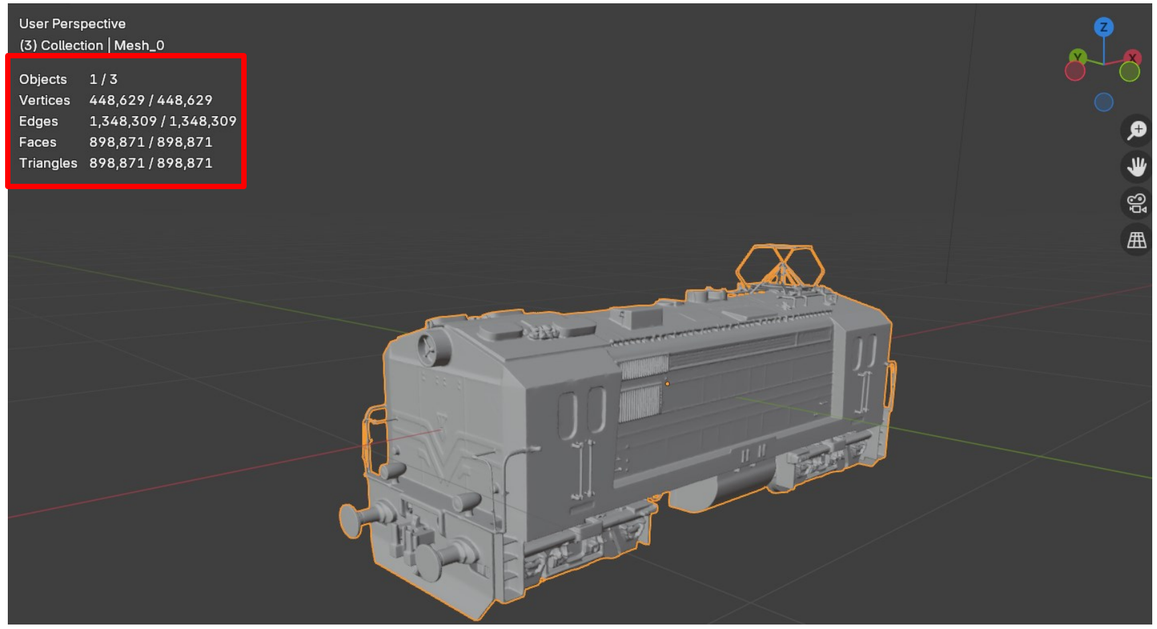


Mesh Density (Different Object)

- ❑ **Task:** Open a New Blender (keep the previous one as well) and Import fbx for Locomotive.fbx and Observe its Mesh Density
- ❑ Compare the two models' vertex, edge, face, and triangle counts.
- ❑ **Think & Observe - No submission required:** Which model has greater mesh density and How might that affect visual detail and computer performance? (If you're unsure, search "mesh density in 3D modeling" before continuing.)



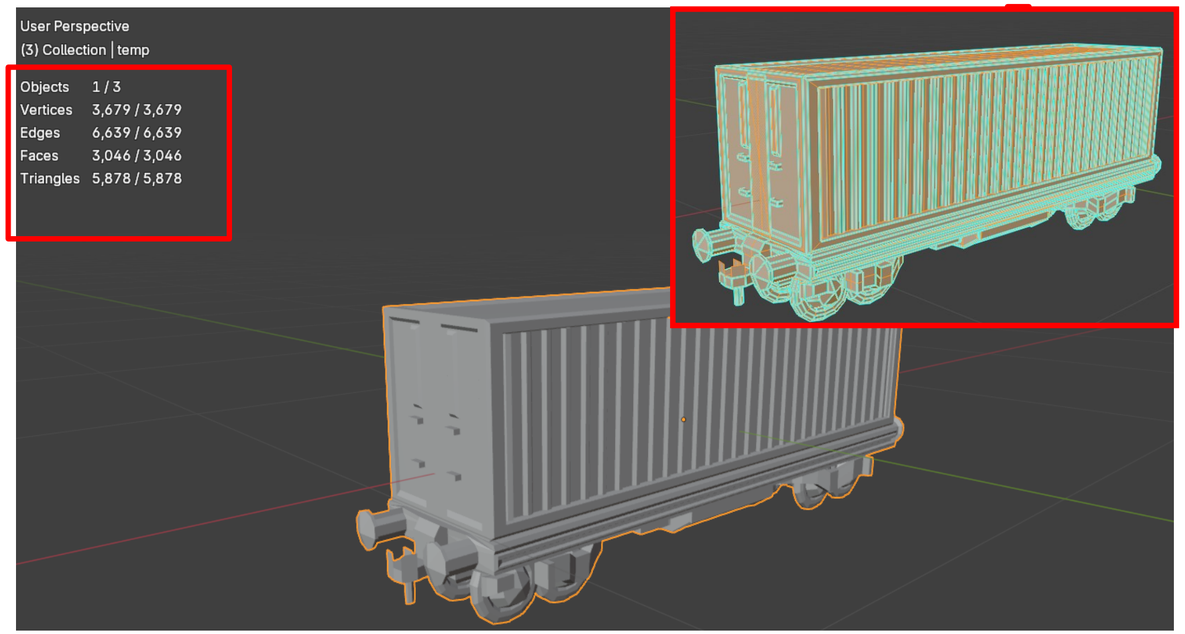
Hint: Make sure you select Locomotive



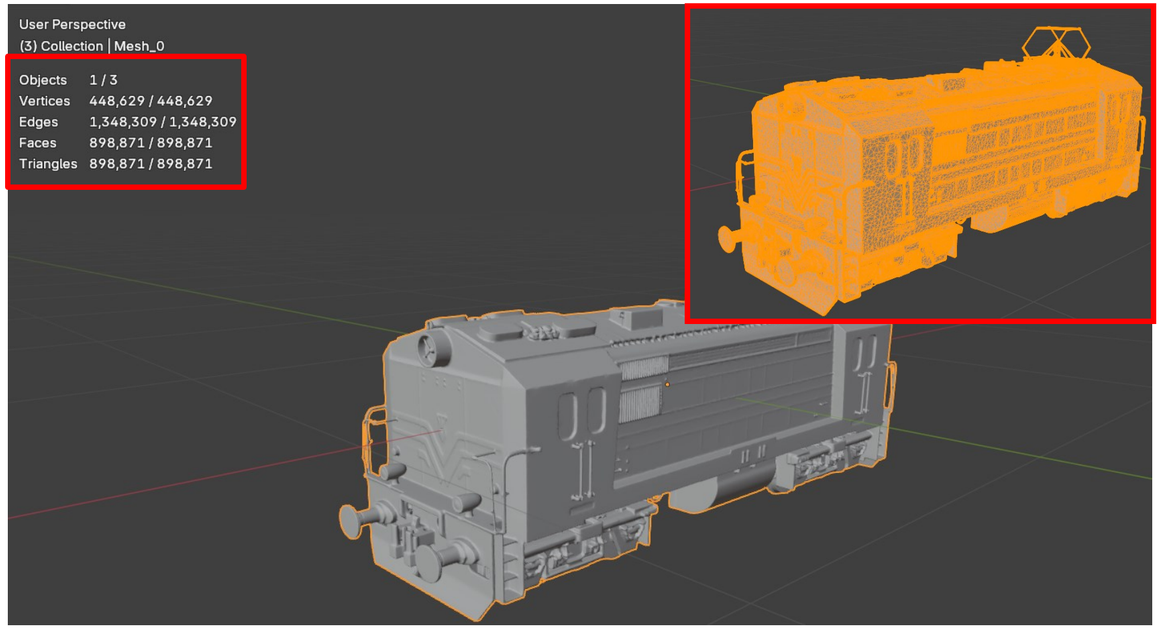
Mesh Density (Different Object)

- **Low-poly** means the object uses a small number of polygons and is optimized for real-time performance.
- **High-poly** means the object has many polygons and more geometric detail, but it may reduce FPS and increase memory usage.
- **Think & Observe - No submission required:** Is there any threshold for number of triangles to determine low/high poly?

Low Poly



High Poly



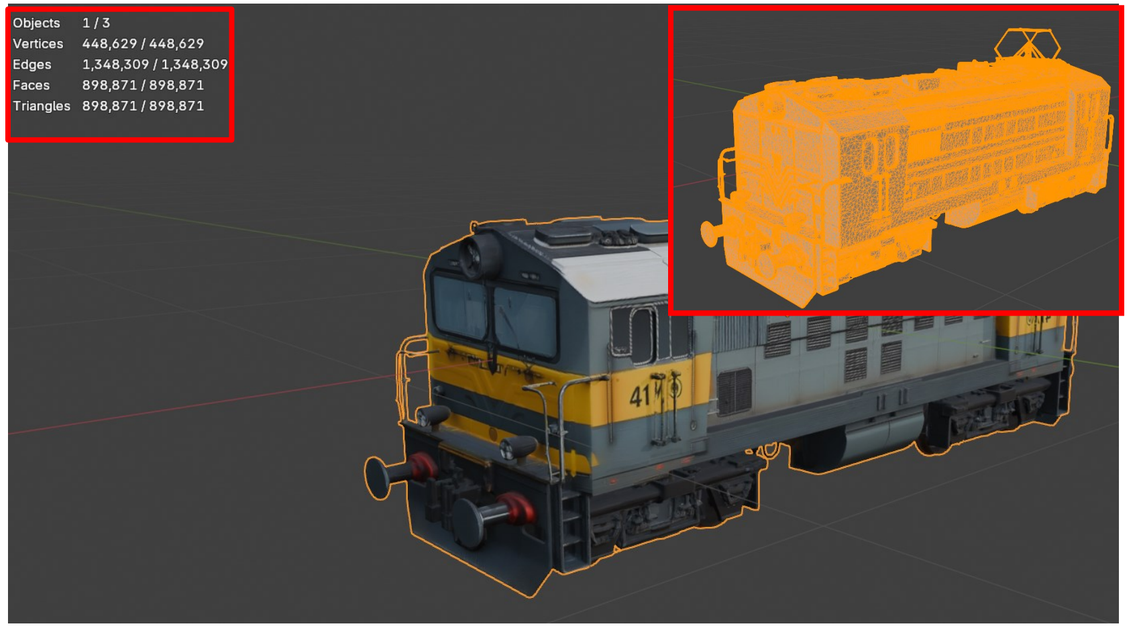
Mesh Density (Same Object)

- ❑ In the first blender → Remove the Wagon and Import LocomotiveV2.fbx
- ❑ Compare the Same Object, different poly
- ❑ Low Poly objects are more optimized and better for PC performance,
- ❑ **Think & Observe - No submission required:** Why might a digital twin developer choose the high-poly version despite its higher performance cost?

Low Poly



High Poly



Always Save the Blender file

- ❑ **Task:** Save the Blender files
- ❑ File → Save as → Put your name (HighPoly.blend) & (LowPoly.blend) → save
- ❑ Look for where you saved it; you should observe your two .blend files **(You do not need to submit these)**
- ❑ Raise Your Hand When You Finished

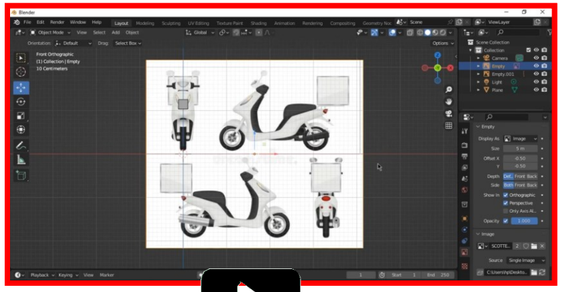
Beginning Of Milestone 1

Develop a Digital Object: Traditional vs Generative AI

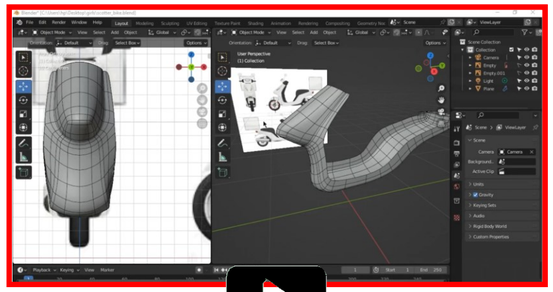
❑ Open the provided link in **Traditional Method** and observe the difference

Traditional Method:
Time Spent: 3 Hours

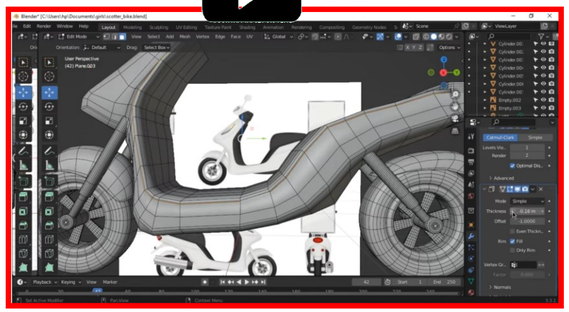
Part 1



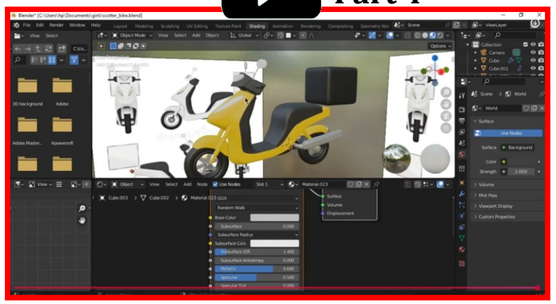
Part 2



Part 3

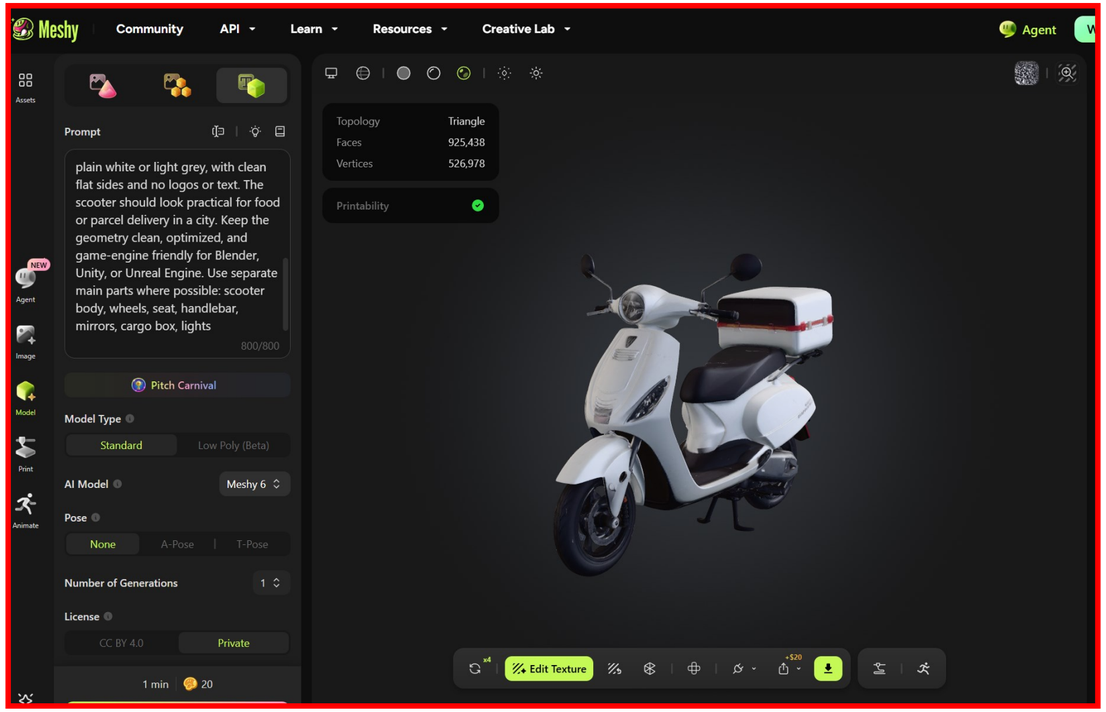


Part 4



https://www.youtube.com/watch?v=TBOAVU5dCbY&t=576s

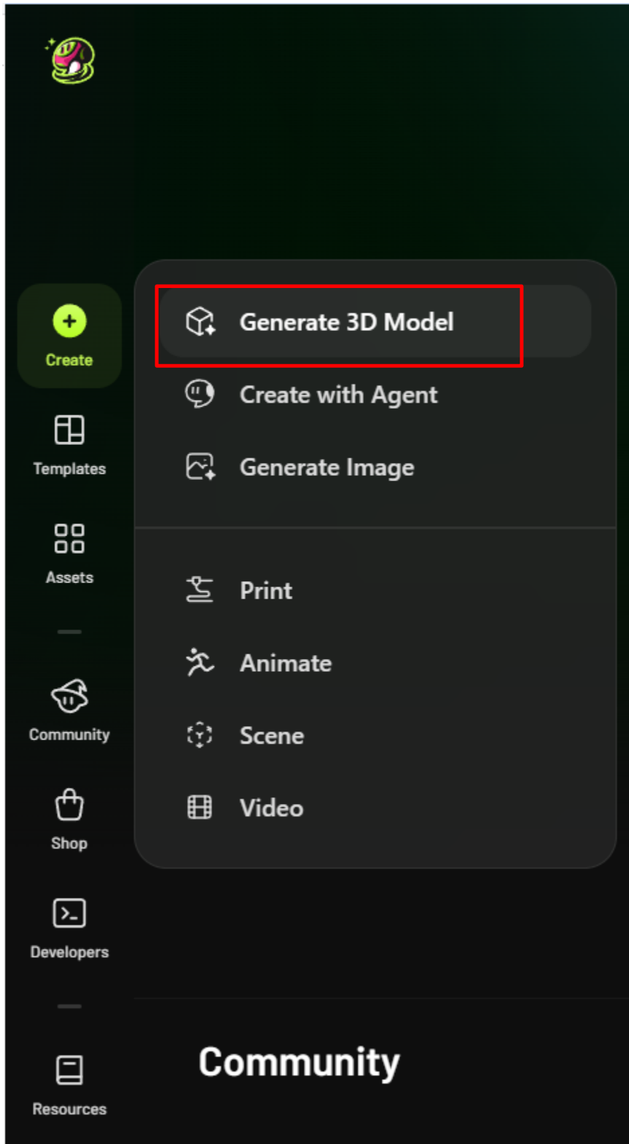
Generative AI
Time Spent: 3 Minutes



https://www.meshy.ai/

Develop a Digital Object using Generative AI

- ❑ Go to www.meshy.ai
- ❑ Create an account
- ❑ YOU **DO NOT NEED TO SUBSCRIBE** FOR THIS EXERCISE
- ❑ Welcome to the 3D modeling (Digital Twin Object Development) world
- ❑ Meshy is an AI-powered generative tool that creates 3D models from reference images
- ❑ Select “Generate 3D Model”



Develop a Digital Object using Generative AI

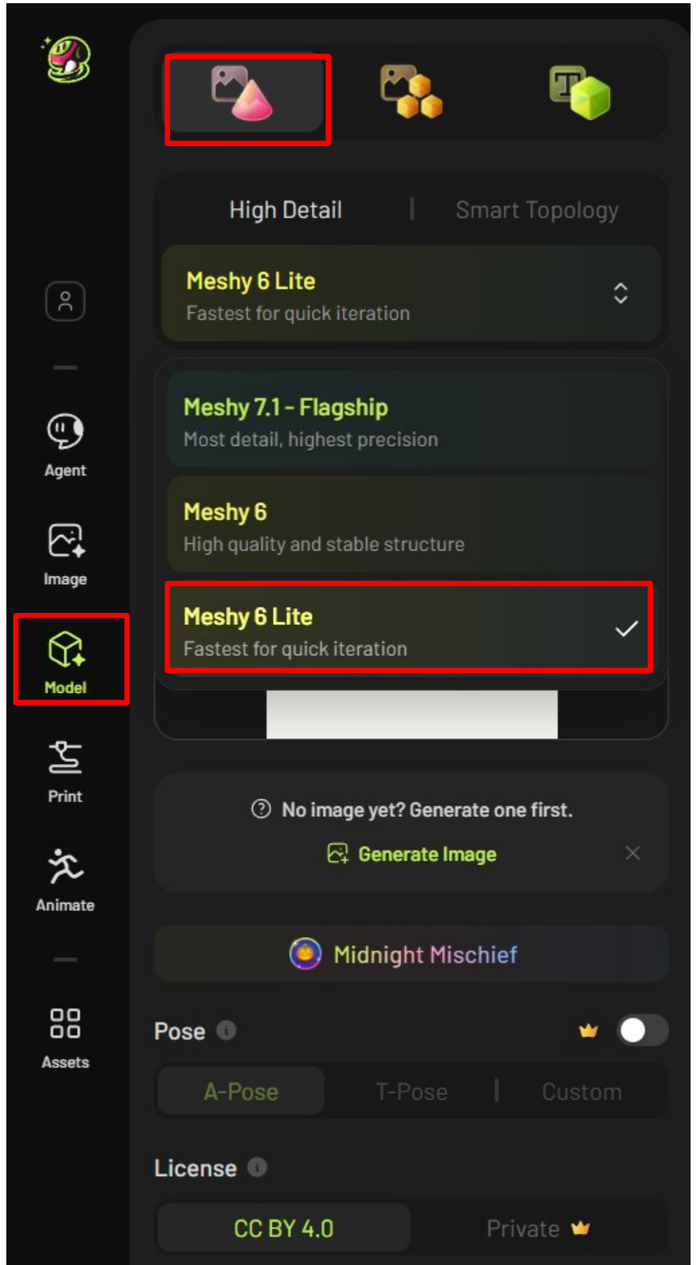
- ❑ Select “Image to 3D”
- ❑ We need to describe what we want in 800 words
- ❑ Ask ChatGPT or any AI to generate 800 words of your object of interest

(a Motorcycle in our case)

For example, I copied and paste an image of interest and asked

“Write a command for Meshy.ai to generate 3D object of this (word limit is 800)”





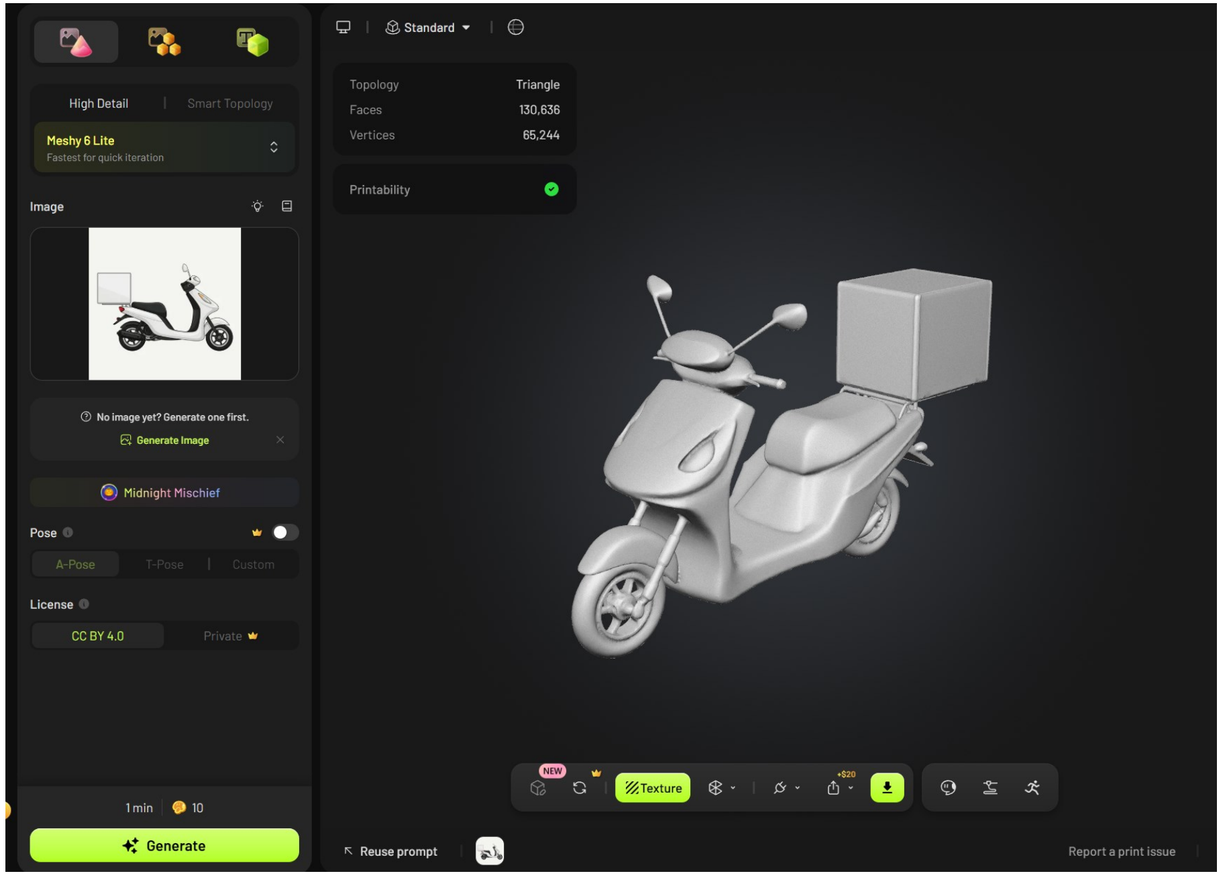
Develop a Digital Object using Generative AI

- ❑ Copy and paste your prompt in Gemini to generate the reference image



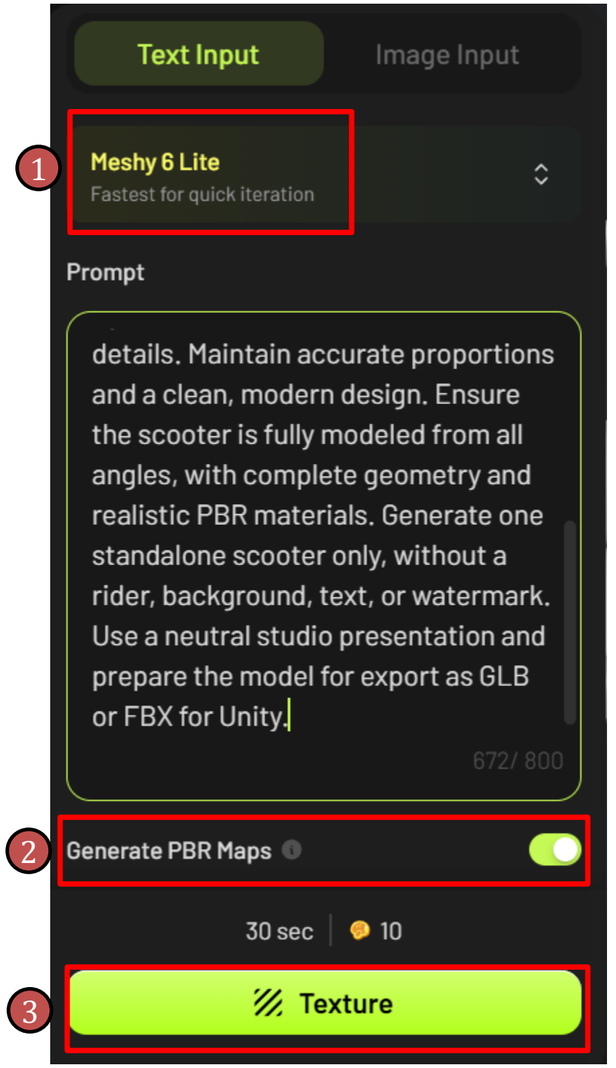
Develop a Digital Object using Generative AI

- ❑ Copy and Paste your generated image in Image box
- ❑ Set the model to High Detail
- ❑ Select Meshy 6 Lite
- ❑ Select “generate”
- ❑ It will consume 10 Tokens (you have **100 free tokens**)
- ❑ Wait until it generate the 3D object (3 Min)
- ❑ See the output
- ❑ Congrats you built your first digital object



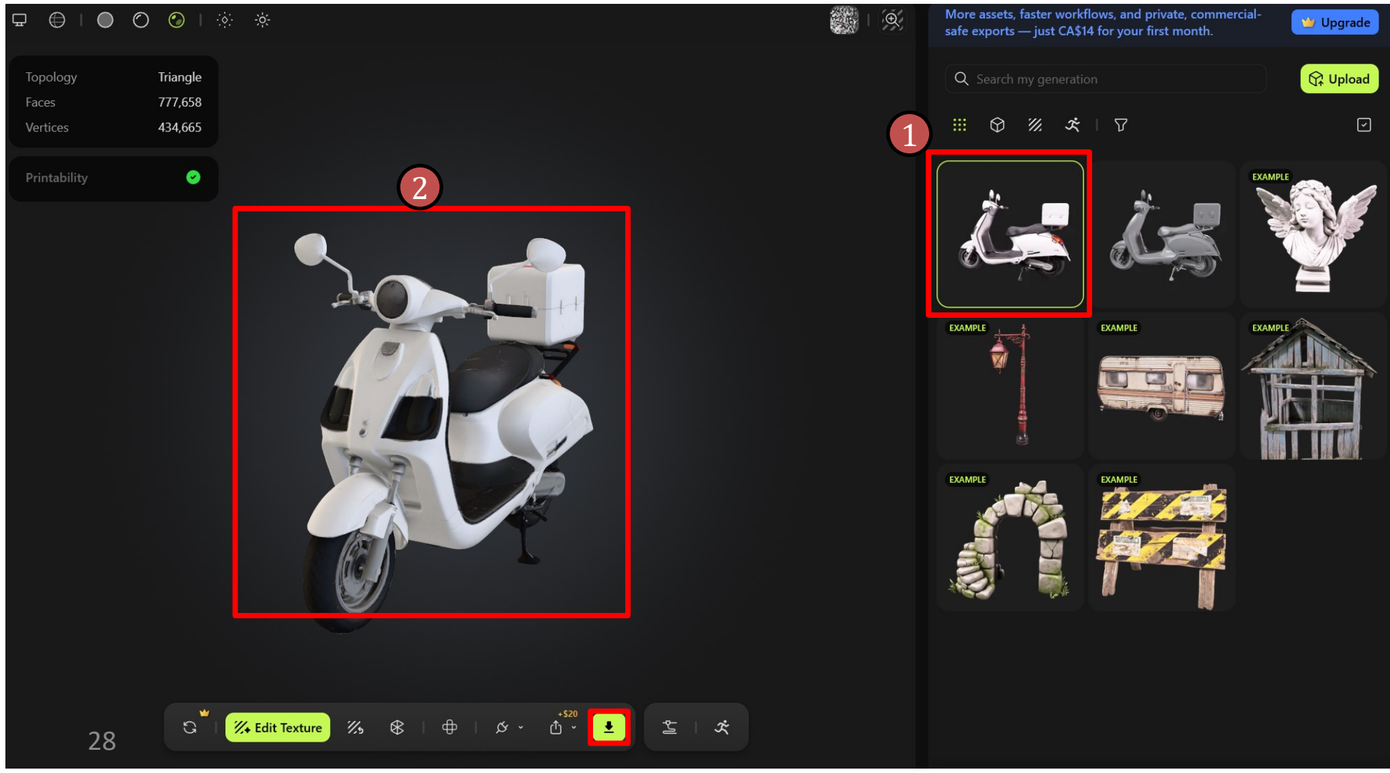
Develop a Digital Object using Generative AI

- ❑ Select “Texture” on the bottom → Your text should be limited to 800 Words
- ❑ If more than 800 words, ask AI to shorten it
- ❑ Uncheck all of the boxes
- ❑ Select “Texture”



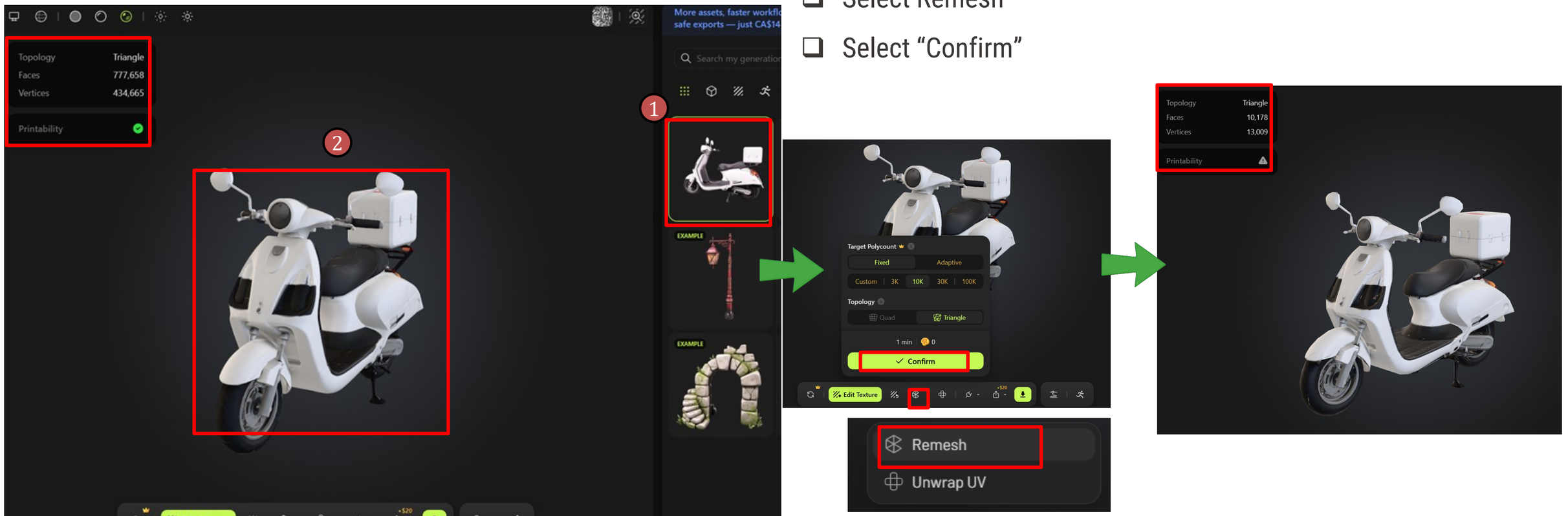
Develop a Digital Object using Generative AI

- ❑ See output
- ❑ Observe number of Faces and Vertices (It is high poly object)
- ❑ Select “Export Button” → You have 10 free downloads (Don’t Download it For Now)
- ❑ See next slide



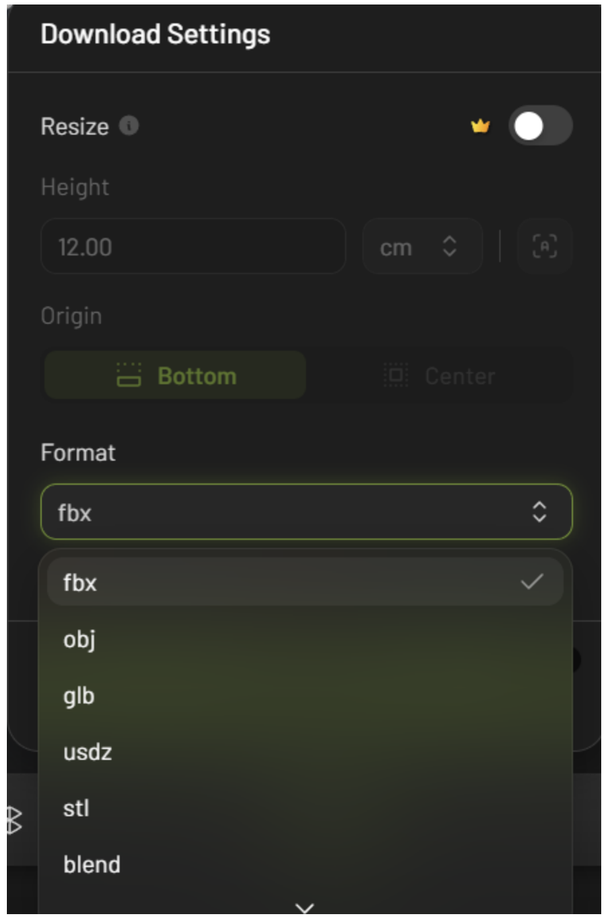
High Poly → Low Poly

- ❑ Observe number of Faces and Vertices
- ❑ **Task:** Regenerate it with reduced number of polygon meshes
- ❑ Select Remesh
- ❑ Select “Confirm”
- ❑ See output
- ❑ Observe number of Faces and Vertices



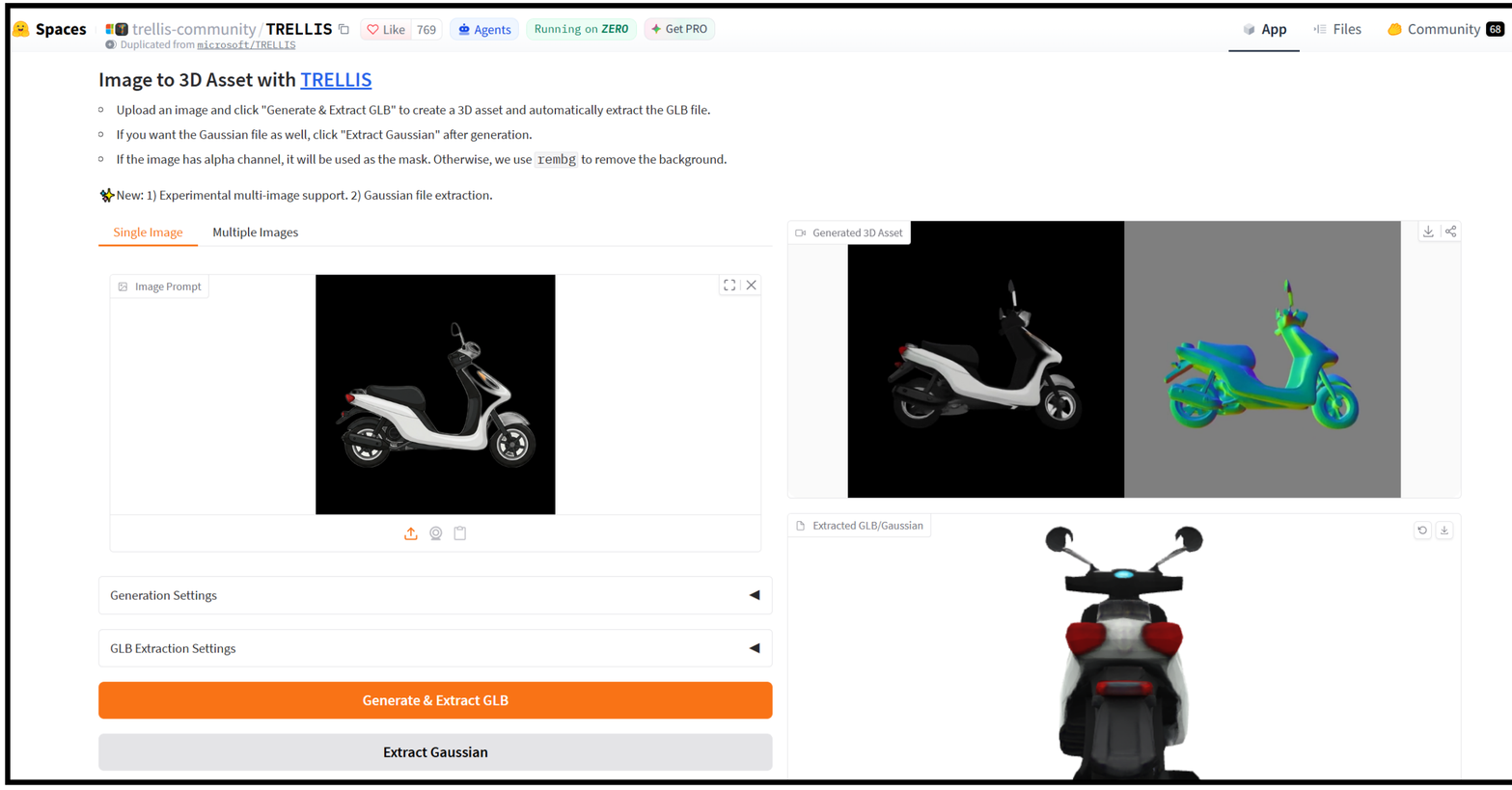
Download 3D Digital Model

- ❑ Download the 3D Digital Model
- ❑ Set the format to FBX
- ❑ Download
- ❑ Import In Blender and See it



Try this Alternative Approach (Open Source)

- Microsoft developed an opensource version of Meshy.ai
- https://huggingface.co/spaces/trellis-community/TRELLIS
- Drag and drop your image and generate GLB (Another format 3D Digital Object)
- Import In **Blender as (.glb)** and See it



Deliverable 1: Develop a Digital Object using Generative AI

- **Task:** By Observing Previous Slides; Generate a Freight Locomotive
- Use **Meshy Version 6 Lite** so you can export the fbx

You may see a different model than this, depends on your command

Sample Output



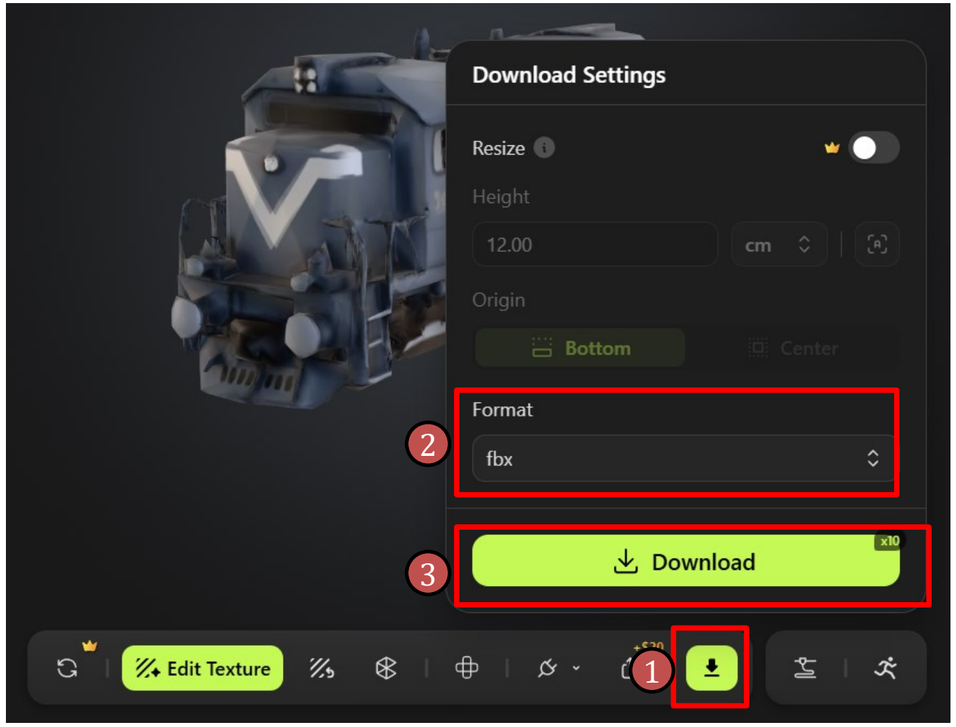
Optimized Sample Output



Deliverable 1: Develop a Digital Object using Generative AI

- ❑ **Task:** Export FBX
- ❑ Change format to FBX
- ❑ Export fbx

Export Settings



Extract the Zip File



Model: .fbx

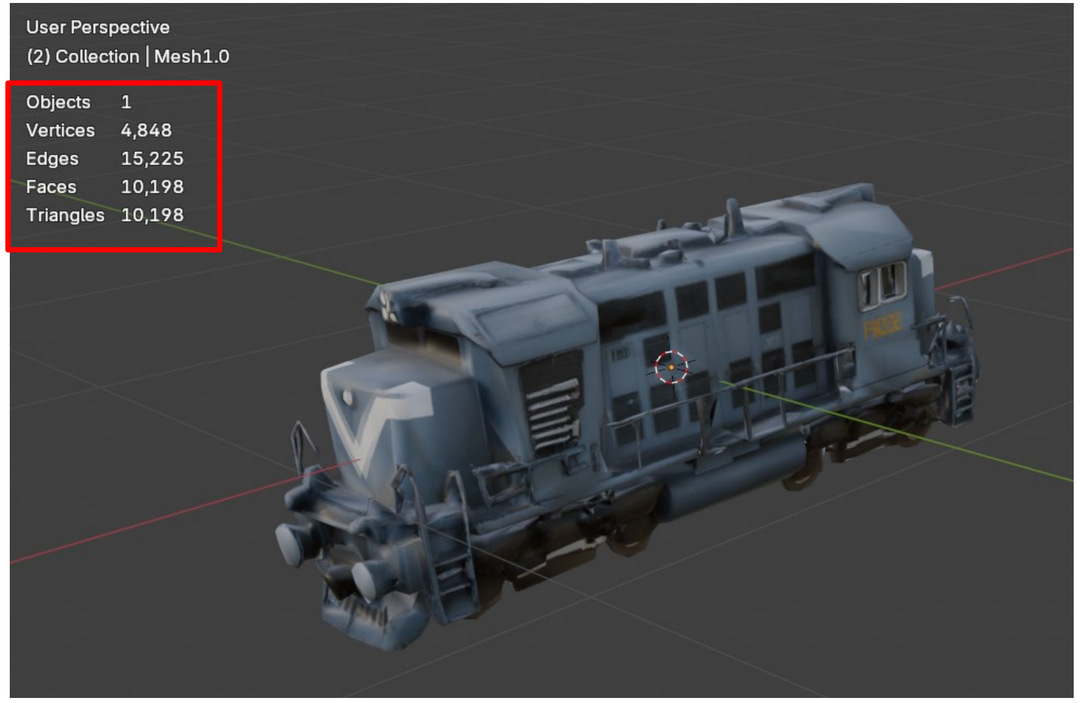
Texture: .png

You may have multiple textures (.png files)

Deliverable 1: Develop a Digital Object using Generative AI

- ❑ **Task:** Import into Blender
- ❑ Open a New Blender
- ❑ Import the Newly Created Model into Blender
- ❑ Observe the number of meshes

Export Settings



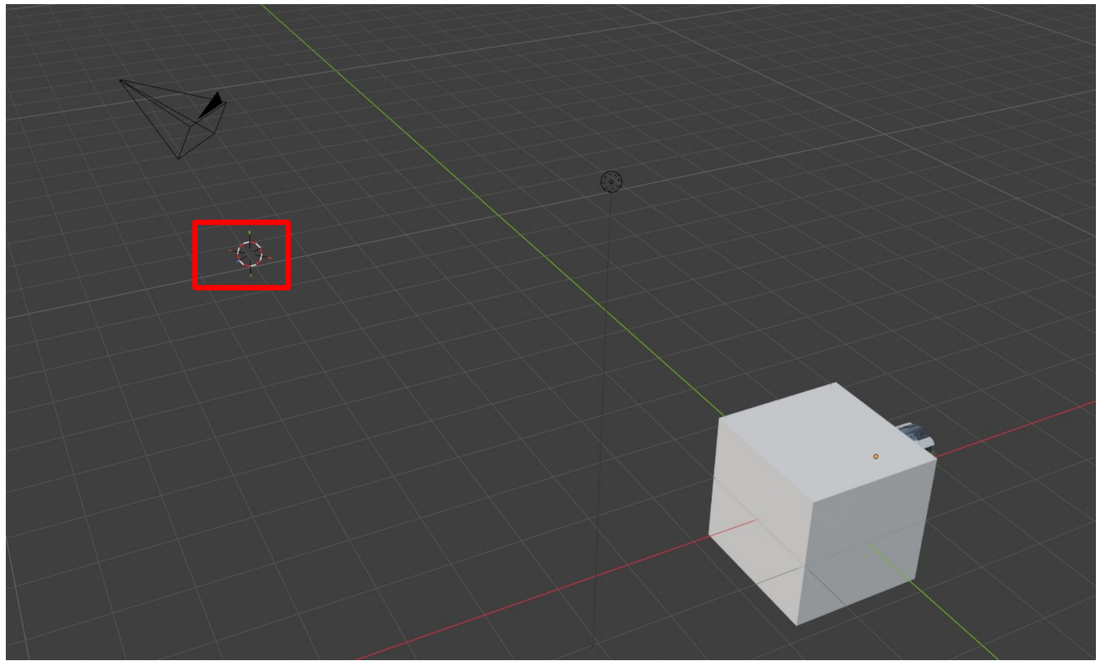
- ❑ **Deliverable (1)**
- ❑ Create a New Folder (FirstLastName)
- ❑ Save your Blender file (Put Your FirstLastName); you will require this for the rest of lab tutorials)
- ❑ Zip the File
- ❑ Submit the created model (At the end of lab)
- ❑ Raise Your Hand When You Finished

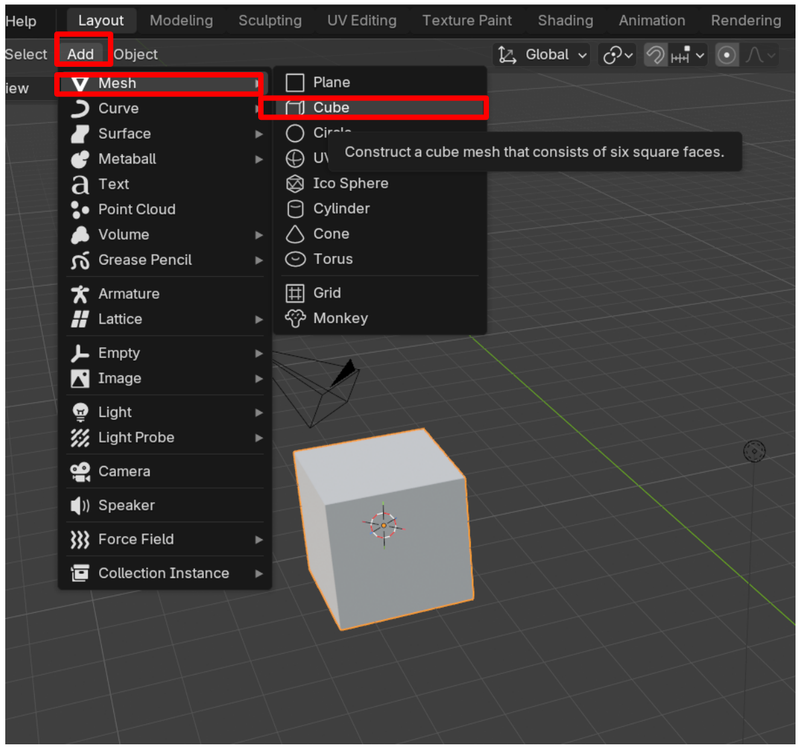
Beginning of Milestone 2

Coordinate Systems and Transform

- ❑ Open a New Blender
- ❑ Select 3D Cursor
- ❑ Select Here
- ❑ **Add Another Cube:** Add → Mesh → Cube

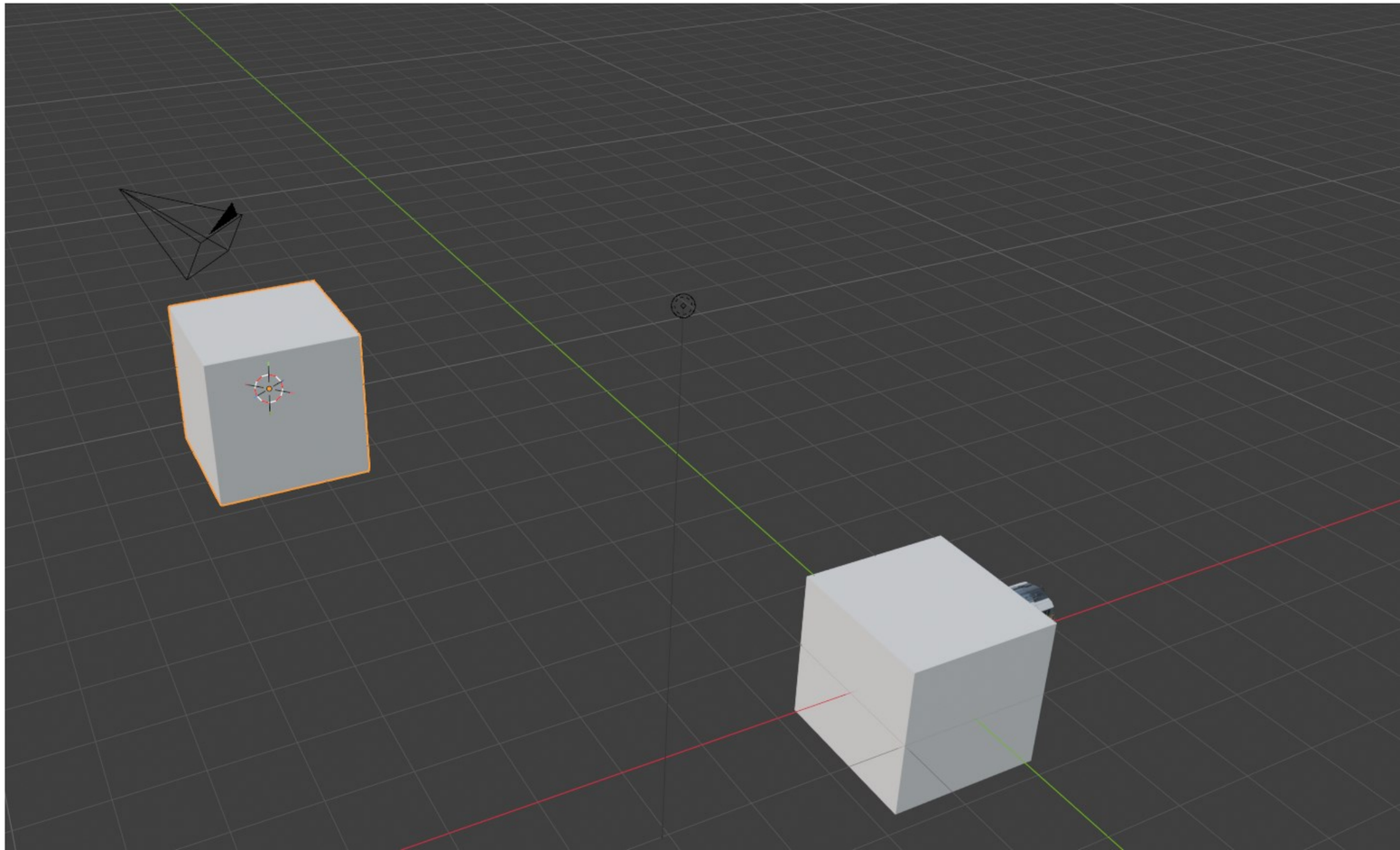






Coordinate Systems and Transform

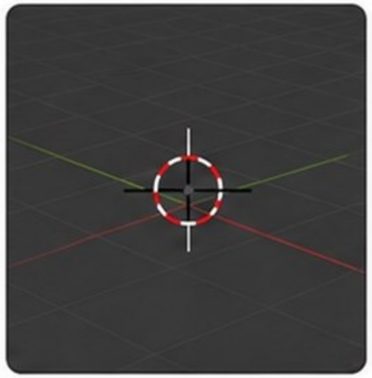
- The new Cube will be Added in the Location of Cursor
- **Note:** Always place cursor in the world origin
- What is cursor? (next slide)



3D Cursor in Blender

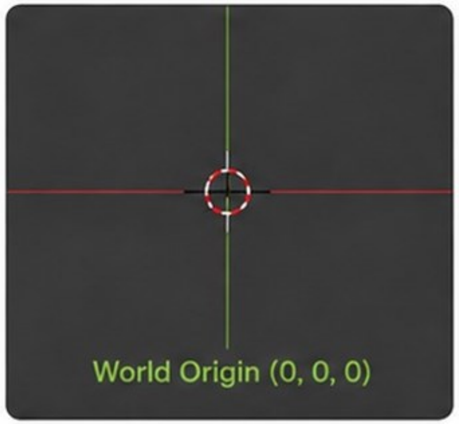
The 3D Cursor is a target point used for adding objects, snapping, modeling tools, and transformations.

What is the 3D Cursor?



- A red–white circle with crosshair in the 3D Viewport
- Represents a precise point in 3D space
- Used as the origin for adding new objects
- Used as the pivot point for many operations

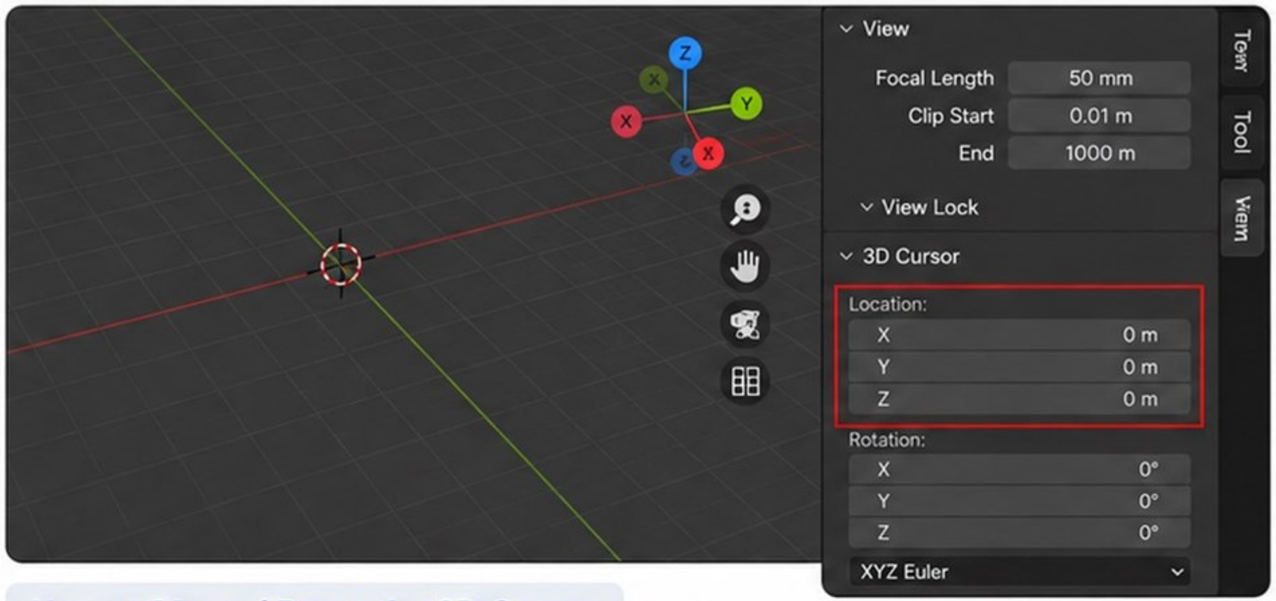
Always Keep the Cursor at the World Origin



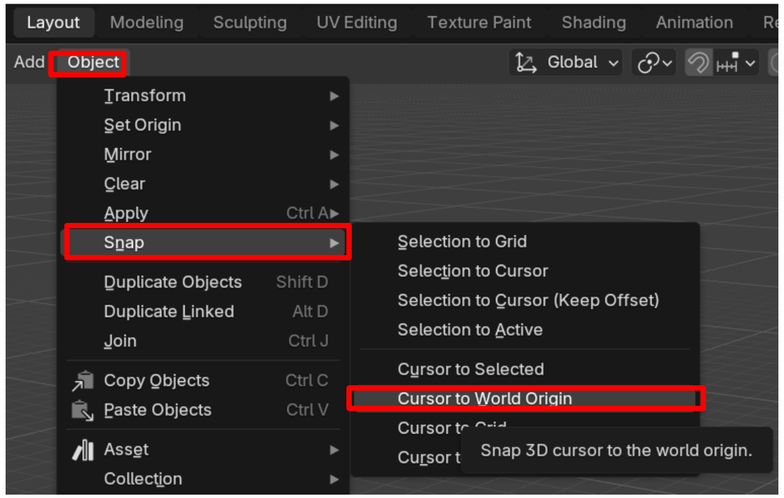
- For a clean and organized workflow, keep the 3D Cursor at the world origin (0, 0, 0).
- This ensures consistency when adding objects, exporting, and working in real-world scale.
- Reset the Cursor to Origin before starting or exporting your scene.

Cursor Location

You can see and edit the cursor location in the **sidebar (N) → View tab.**



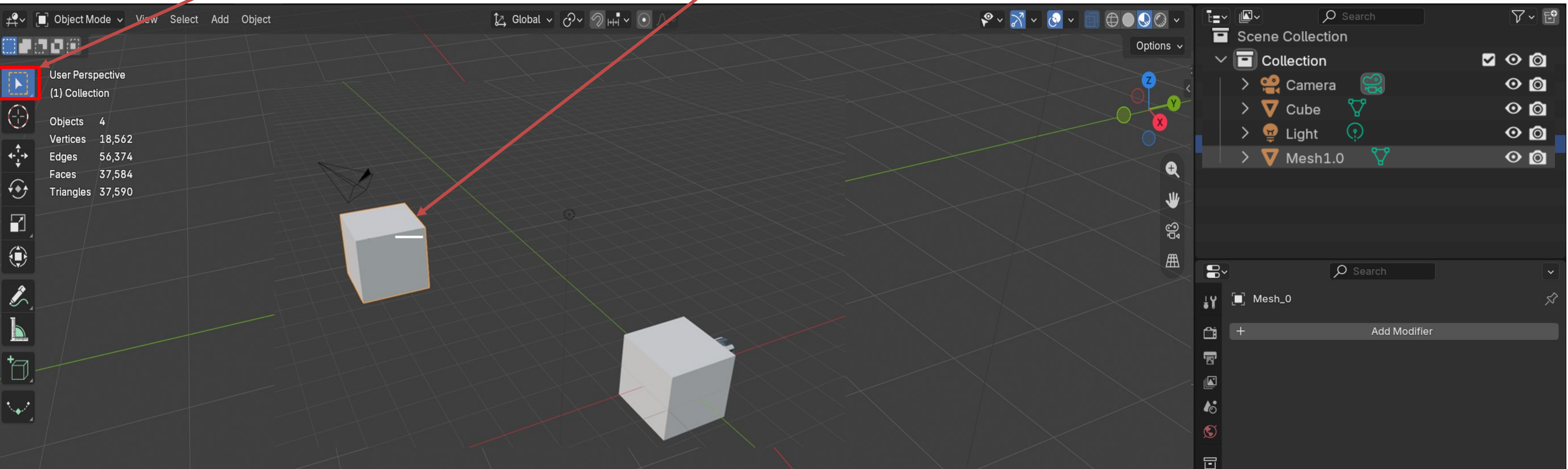
How to Move / Reset the 3D Cursor



Do This

Coordinate Systems and Transform

- ❑ **Remove the newly added Cube:** Select “Select Box”, then select the second cube → Press “Delete”
- ❑ Import LocomotiveV2.fbx
- ❑ That Object is Small inside the Cube

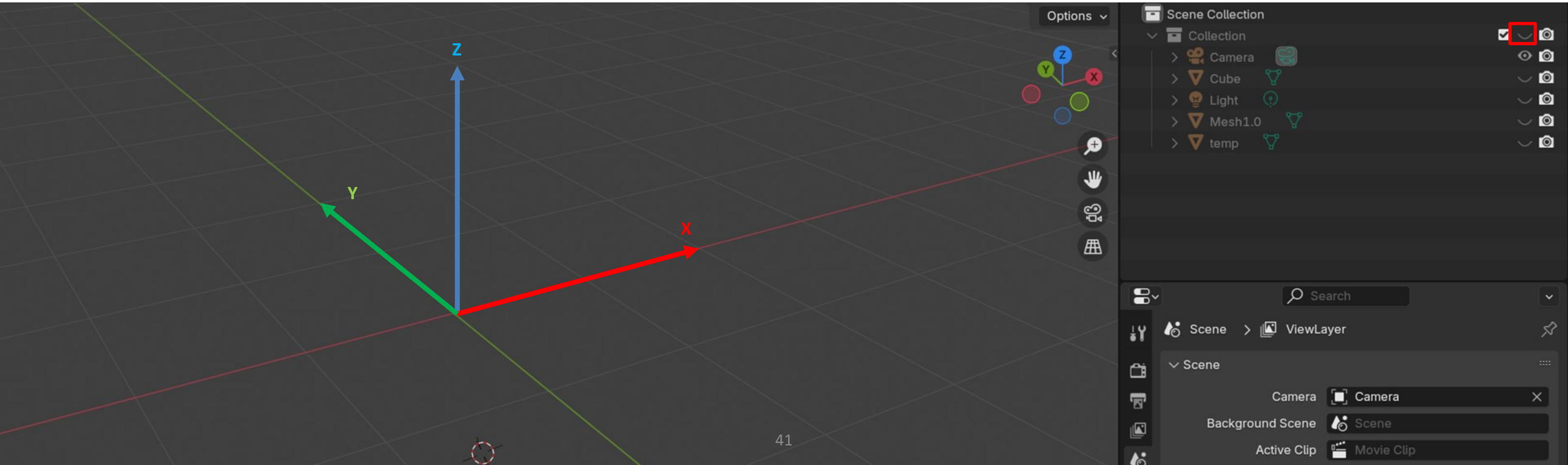


Coordinate Systems and Transform

- ❑ Save your blender File

Coordinate Systems and Transform

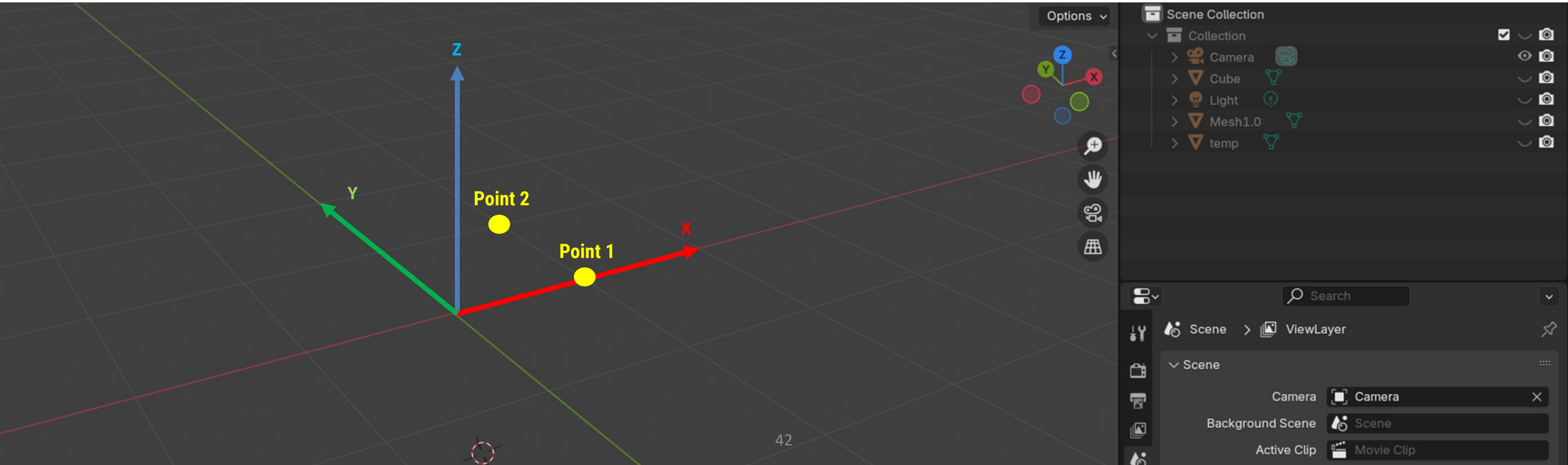
- ❑ Hide all objects
- ❑ Observe the coordinate system in the blender origin point (I drew this-you don't need to)



Coordinate Systems and Transform

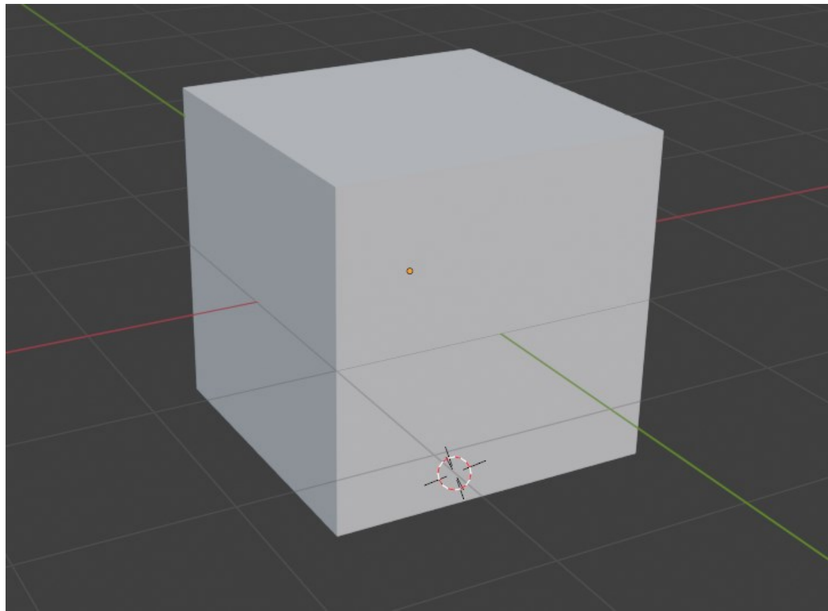
- ❑ **Think & Observe - No submission required:**
- ❑ What are the coordinates of point 2?
- ❑ **Response:**
- ❑ (1,0,0)

- ❑ **Think & Observe - No submission required:**
- ❑ What are the coordinates of point 2?
- ❑ **Response:**
- ❑ (1,1,0)



Coordinate Systems and Transform

- ❑ **Unhide Cube**
- ❑ What are the cube’s dimensions (length, width, and height)?
- ❑ **Hint:** Select the cube and check Dimensions in the Transform panel.
- ❑ **Think & Observe:** Assume this cube is imported into a VR scene at its current dimensions and placed directly in front of the user. The cube is opaque and cannot be moved. Which statement best describes the user’s view??
- ❑ A) The user can hold the cube in one hand.
- ❑ B) The cube may block the user’s view of objects directly behind it.
- ❑ C) The user can clearly see all objects behind the cube.
- ❑ D) The cube’s dimensions automatically change when it is imported into VR.



- ❑ **Correct Response:** The cube may block the user’s view of objects directly behind it.

Coordinate Systems and Transform

- ❑ **Hide all Objects**
- ❑ **Unhide Mesh1.0** (LocomotiveV2.fbx) → Low Poly Locomotive



Always Apply Transform After Importing an Object

- ❑ Select the object (e.g., Mesh1.0) in the Outliner or in the 3D view.
- ❑ Go to **Object → Apply → All Transforms**.
- ❑ This sets **Location = 0, Rotation = 0°, and Scale = 1** (applies the current transform).

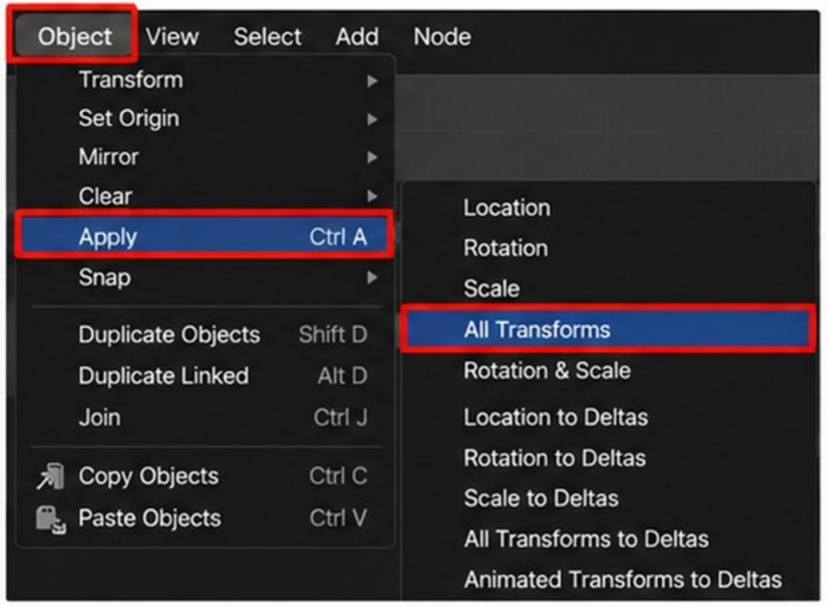
1 Select the object

Choose the object in the Outliner or in the 3D view (e.g., Mesh1.0).



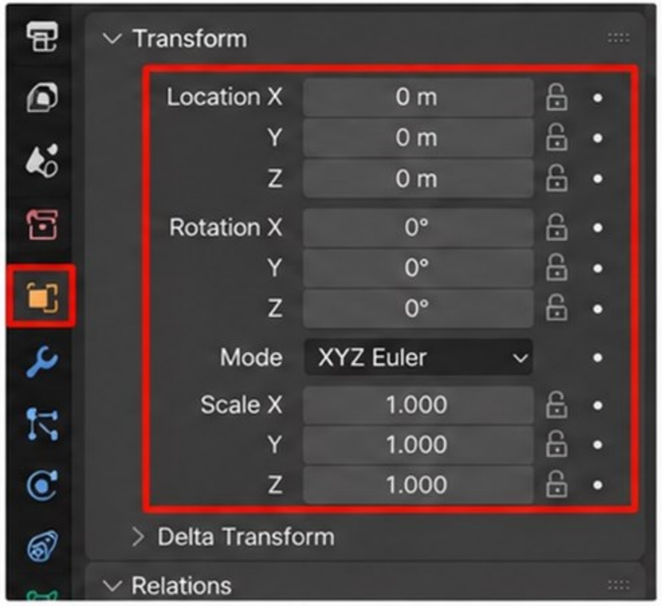
2 Apply all transforms

In the top menu, go to Object → Apply → All Transforms.



3 Result

Location values become 0, rotation becomes 0°, and scale becomes 1.

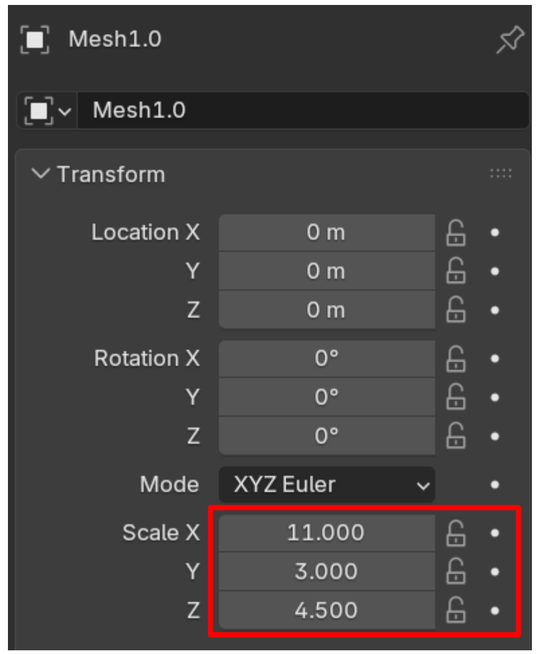


Key idea: Applying all transforms keeps the model's current position, rotation, and size, but resets the transform values to standard values (Location = 0, Rotation = 0°, Scale = 1). This avoids scaling or positioning issues when importing into Unity.

Coordinate Systems and Transform: Scale

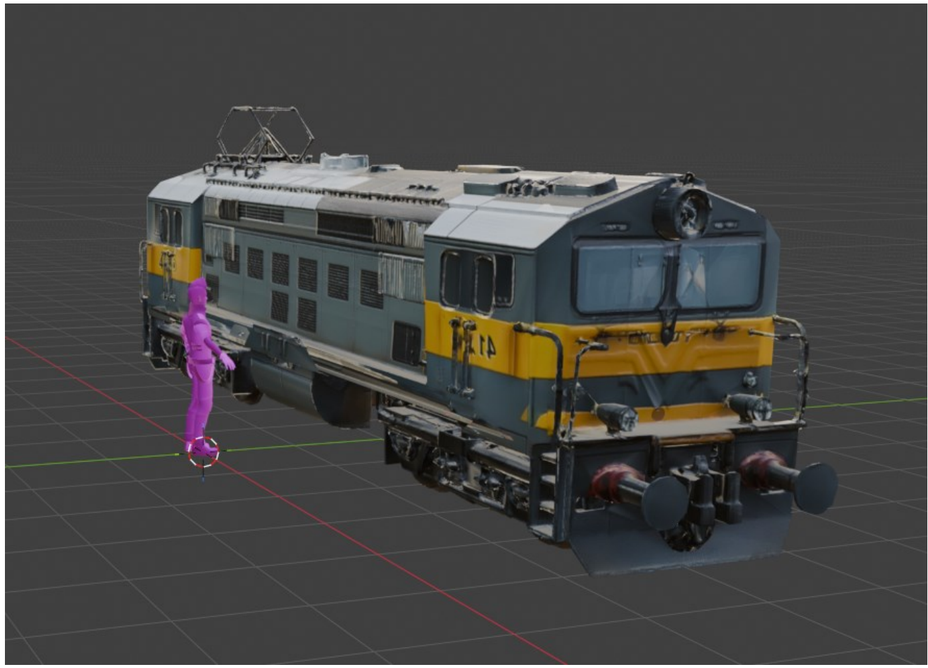
- ❑ **Task:** Resize the Train
- ❑ Search in Google “whats the dimension of a freight locomotive”
- ❑ Assume Length: 11m, Width: 3m, Height: 4.5 → Adjust those values





Coordinate Systems and Transform: Scale

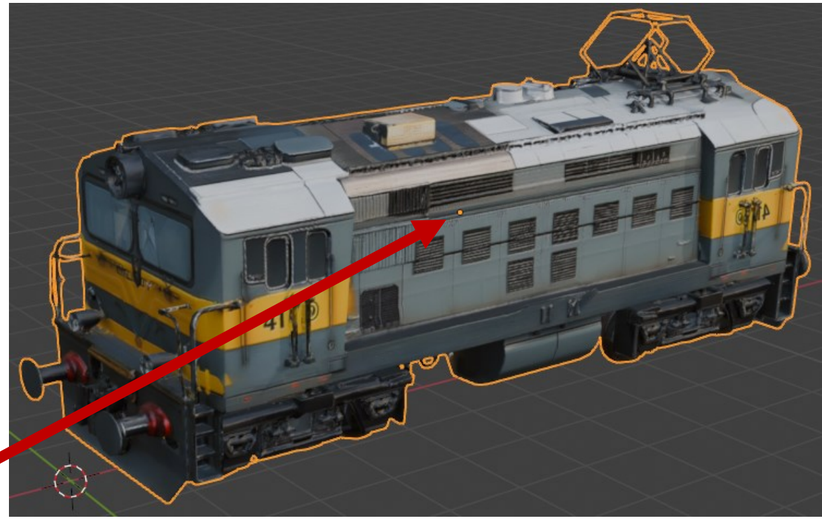
- ❑ The locomotive appears stretched and its proportions are unrealistic.
- ❑ Undo the previous scaling changes to restore the original proportions.
- ❑ Import Human.fbx as a human reference model with realistic dimensions.
- ❑ Use the human model as a scale reference to adjust the locomotive's dimensions.
- ❑ Ensure the locomotive maintains realistic proportions relative to the human model.

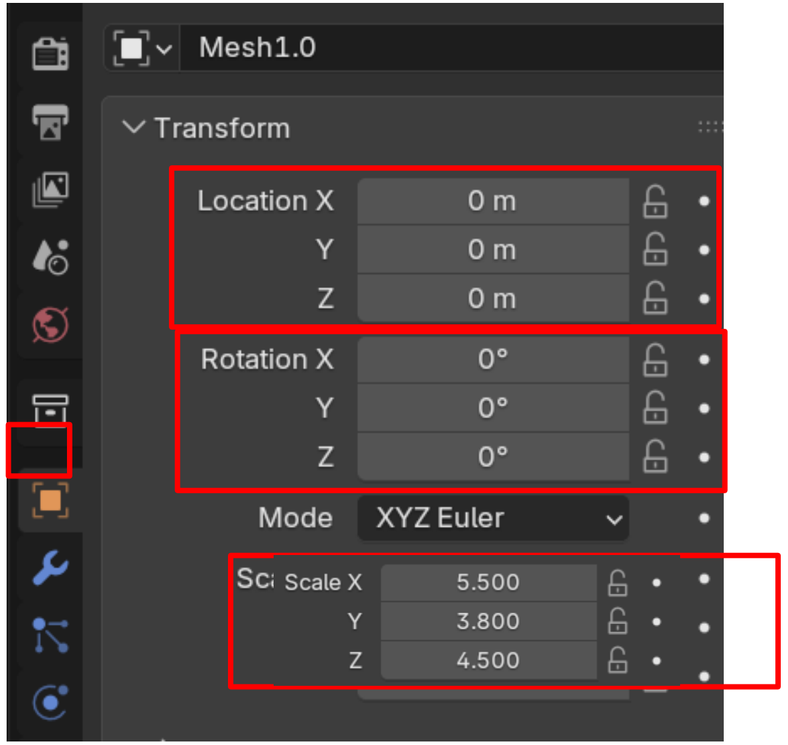


Coordinate Systems and Transform

- ❑ **Task:** Resize the Train
- ❑ Select Train
- ❑ Select “Object Properties” including Location, Rotation, Scale

Property	Meaning	Based on what point?
Location	Where the object is placed in the scene	Based on the **object origin**
Rotation	How the object is turned around X, Y, Z axes	Rotates around the **object origin**
Scale	How much the object is enlarged or reduced	Scales from the **object origin**





Coordinate Systems and Transform: Coordinate Alignment

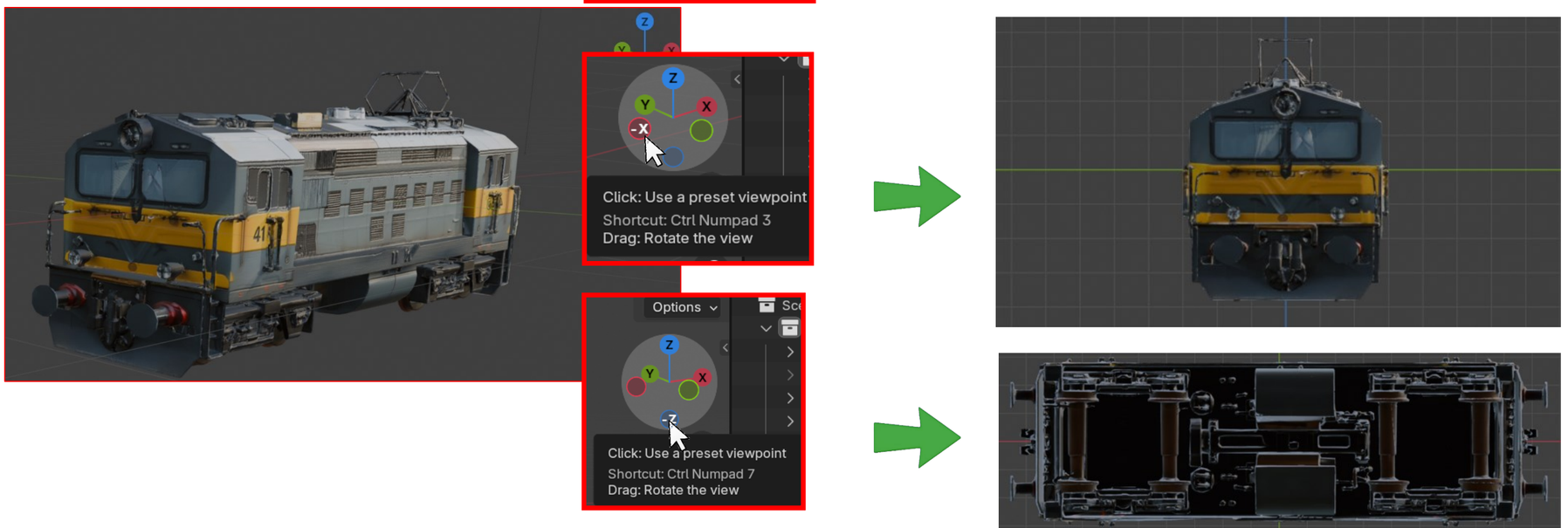
Coordinate Alignment Requirement

- ❑ Adjust the train model (**for instruction, see next slide)** using the X, Y, Z coordinate system
- ❑ The wheels must touch the ground plane (Z = 0).
- ❑ The front of the train should face the Y-axis direction.
- ❑ The train should be centered and symmetrical around the vertical Z-axis.



Coordinate Systems and Transform: Coordinate Alignment

- ❑ **Task:** View the Object along One Coordinate Perspective
- ❑ View it along Y axis
- ❑ View it along X axis
- ❑ View it along Z axis



Coordinate Systems and Transform: Coordinate Alignment

- ❑ **Task:** Move it along Z Axis
- ❑ Select –Y
- ❑ Select Object
- ❑ Press “G”, then Press “Z”

- ❑ Move it until exactly the wheel touches the X Axis

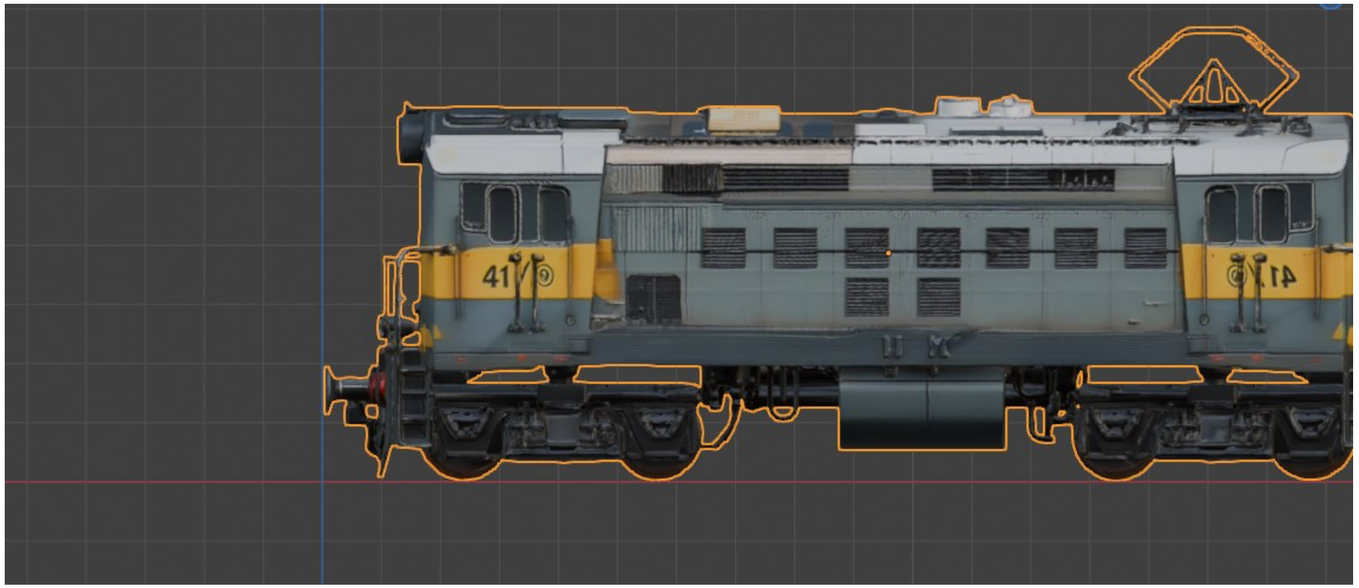




Coordinate Systems and Transform: Coordinate Alignment

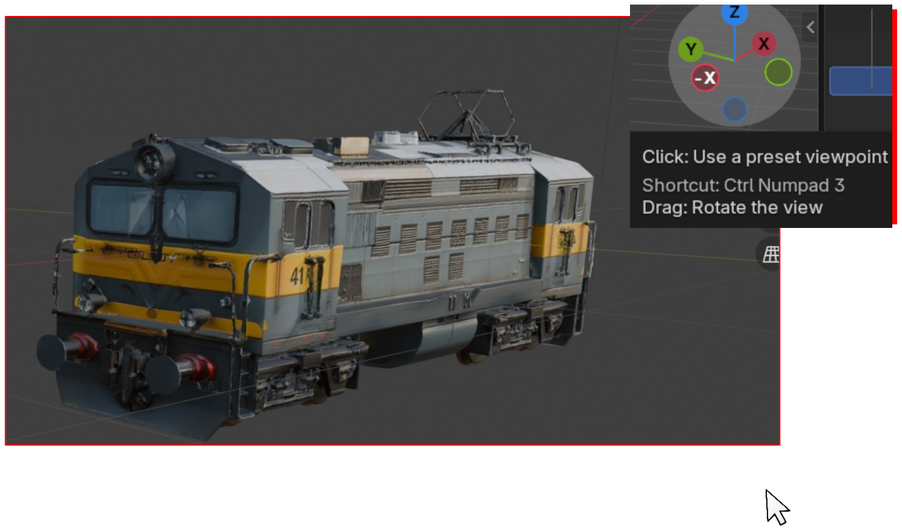
- ❑ **Task:** Move it along X Axis
- ❑ Select –Y
- ❑ Select Object
- ❑ Press “G”, then Press “X”
- ❑ Move it until exactly the front train touches the Z Axis





Coordinate Systems and Transform: Coordinate Alignment

- **Task:** Move it along Y Axis
- Select –X
- Select Object
- Press “G”, then Press “Y”



- Move it until the Z axis exactly divide train by two



Why Location / Rotation / Scale Should Be Clean Before Export

Best practice before exporting from Blender to Unity, Unreal, or a digital twin platform

unity | UNREAL ENGINE | DIGITAL TWIN

Recommended Transform Values for a Single Asset

Location	0, 0, 0
Rotation	0, 0, 0
Scale	1, 1, 1

These values make the model clean, predictable, and easier to use after export.

BEFORE (Not Clean)

Train

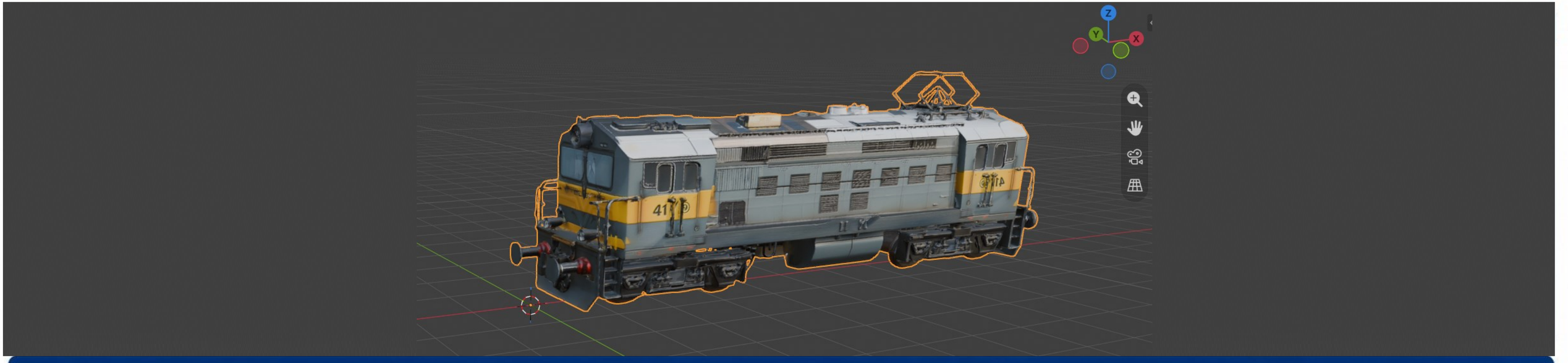
Location	m	
X	5.528 m	Not at origin
Y	0 m	
Z	2.248 m	
Rotation	°	
X	-0.000003°	Rotation is not 0
Y	0°	
Z	90°	
Scale		
X	5.776	Scale is not 1
Y	5.776	
Z	5.776	

Object → Apply → All Transforms

AFTER (Clean)

Train

Location	m	
X	0 m	At origin
Y	0 m	
Z	0 m	
Rotation	°	
X	0°	Rotation is 0
Y	0°	
Z	0°	
Scale		
X	1.000	Scale is 1
Y	1.000	
Z	1.000	



Best Practice: Before export, use **Object → Apply → All Transforms.**

For single assets, clean transforms are preferred.
For full scenes, location may be kept if layout is important.

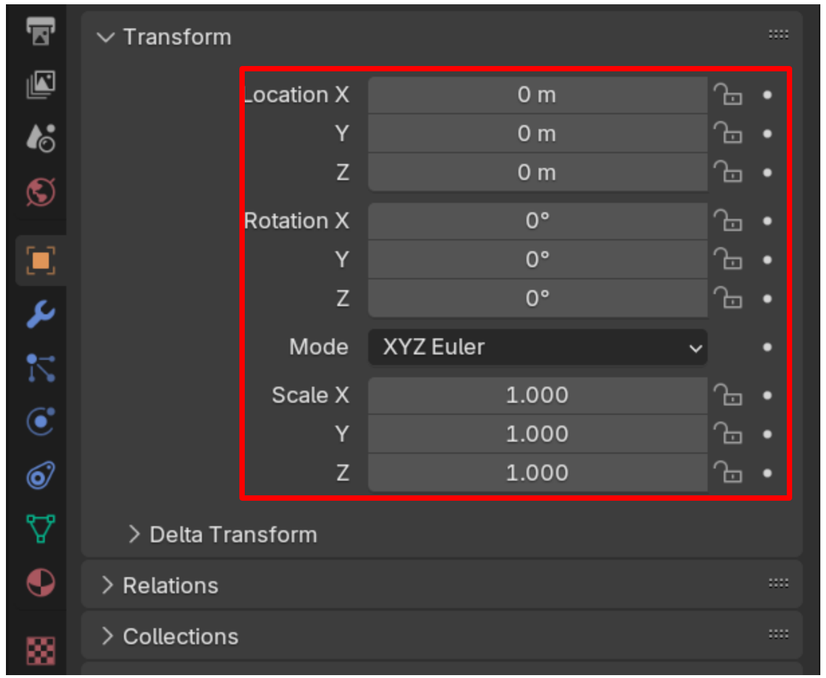
Apply All Transforms

- Object → Apply → All Transform
- It should look like below

3D Cursor in the world Origin

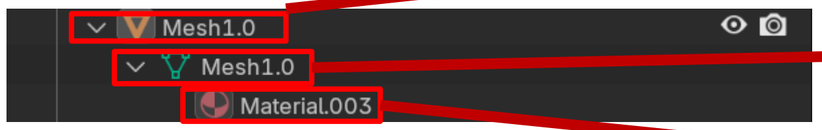


Cleaning Location, Rotation, Scale



Object Main Parts

- **Task:** Select drop down button



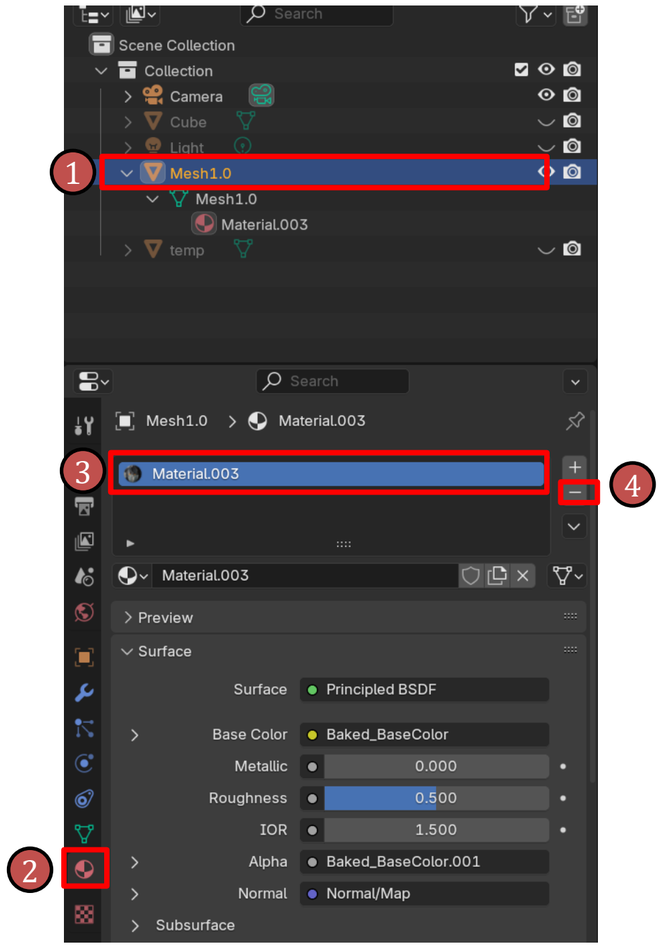
This is the whole object in the scene. It stores the object's Location, Rotation, Scale, visibility, and position in the scene.

This is the actual geometry of the object: vertices, edges, faces, and shape. It defines what the object looks like physically.

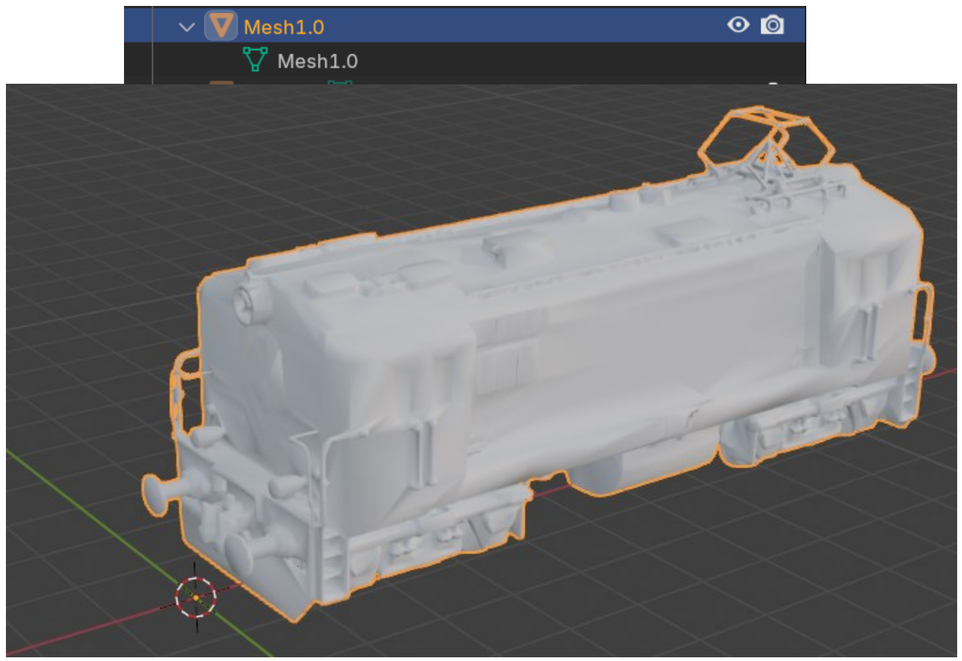
This controls the surface appearance: material, texture.

Materials and Texture

- ❑ Select object
- ❑ Select Material Properties

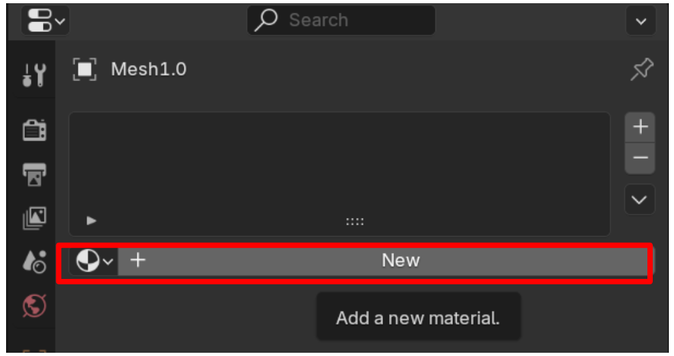


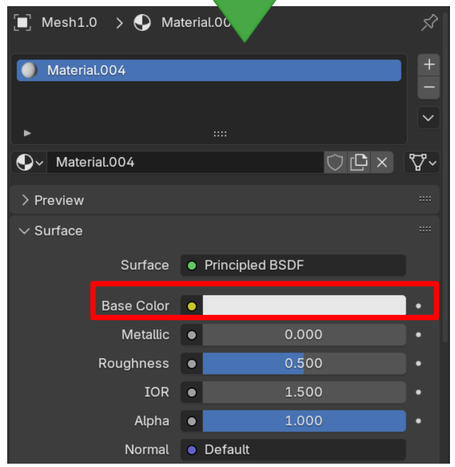
- ❑ Select Material.003
- ❑ Remove the Material.003 by selecting “_”
- ❑ The train will not have any Material



Materials

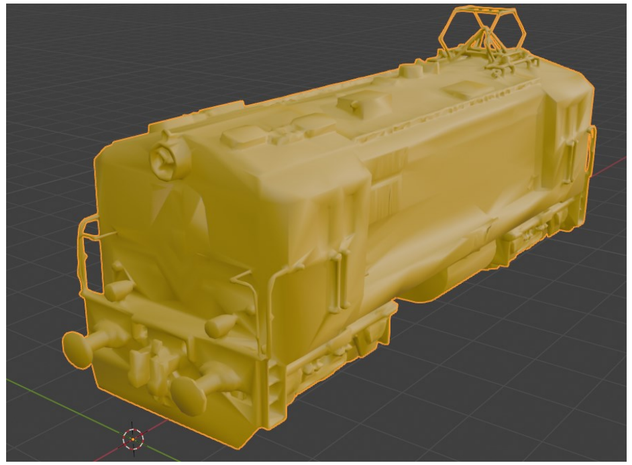
- Add a New Material by Selecting “New”





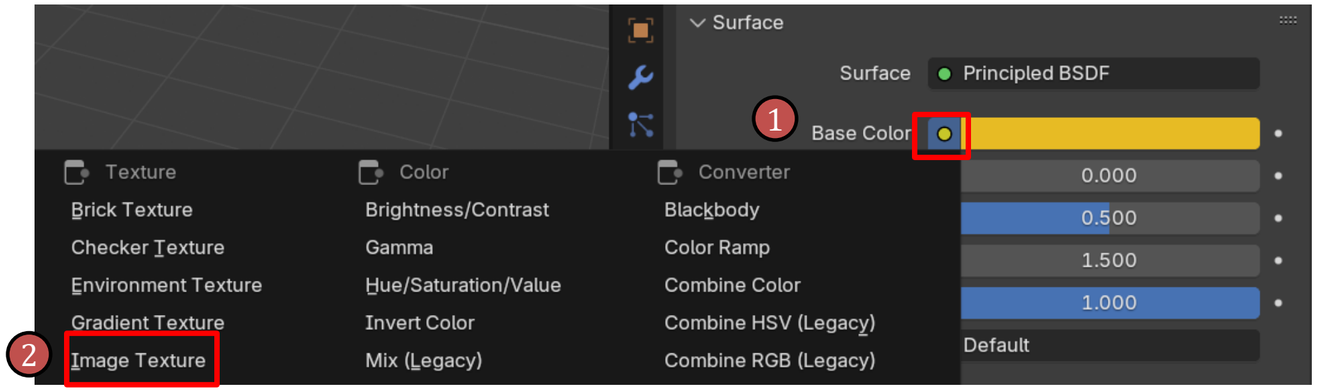
Select Base Color and Pick Yellow Color

The Train has now a Yellow Color assigning to its Material

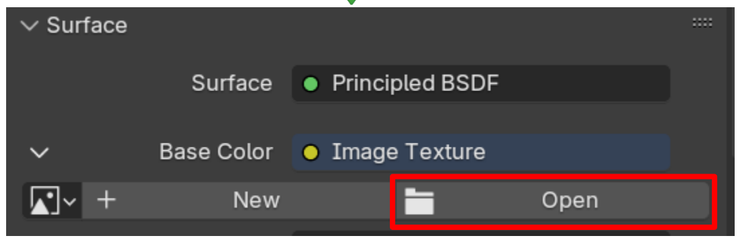


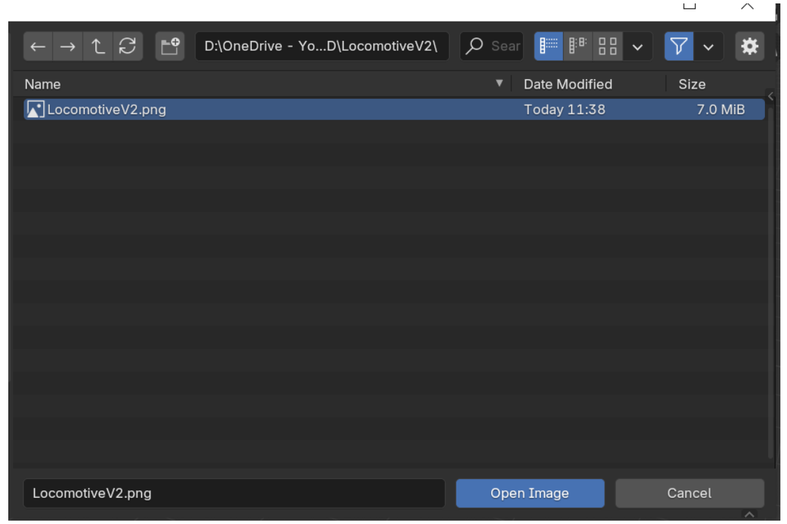
Texture

- Assign texture instead of color to Material









Think & Observe - No submission required

What is the main difference between a material, a color, and a texture in 3D modeling?

A. A material defines how a surface interacts with light, a color is one visual property, and a texture adds image-based or procedural surface detail.

B. A material changes an object’s geometry, a color changes its size, and a texture changes its position.

C. Materials, colors, and textures are different names for exactly the same property.

D. A texture controls lighting, a material only adds images, and a color changes the object’s mesh density.

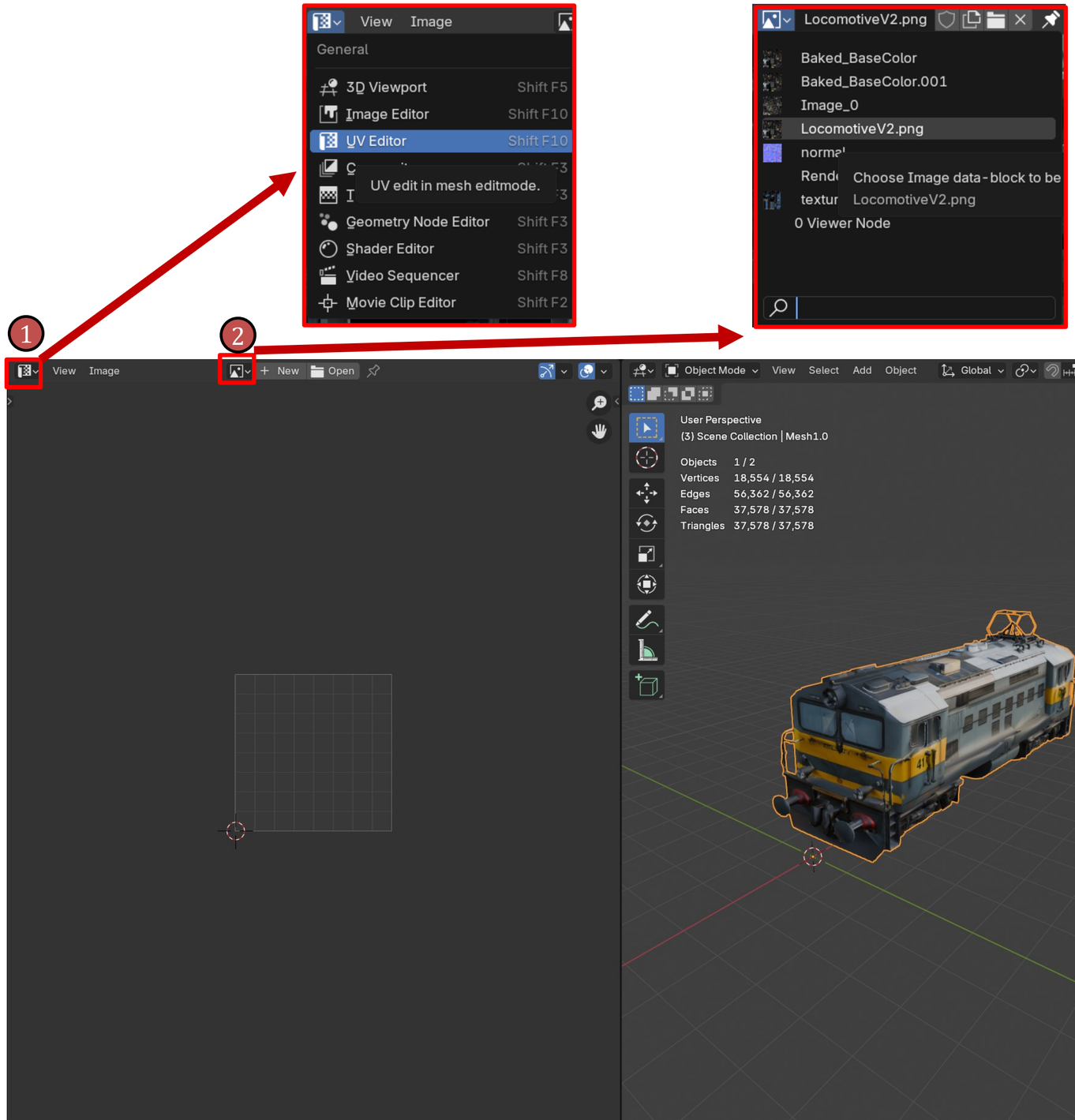
Response: Option A

UV Mapping

- ❑ Split the display into two (Search in the internet or AI how to do it)
- ❑ Select General → UV Editor 1
- ❑ Select “Browse Image to be linked” →2
- ❑ Select LocomotiveV2.png

- ❑ Select Train Object in Viewport
- ❑ Change Object Mode to Editor Mode 3
- ❑ Select “face selection mode”

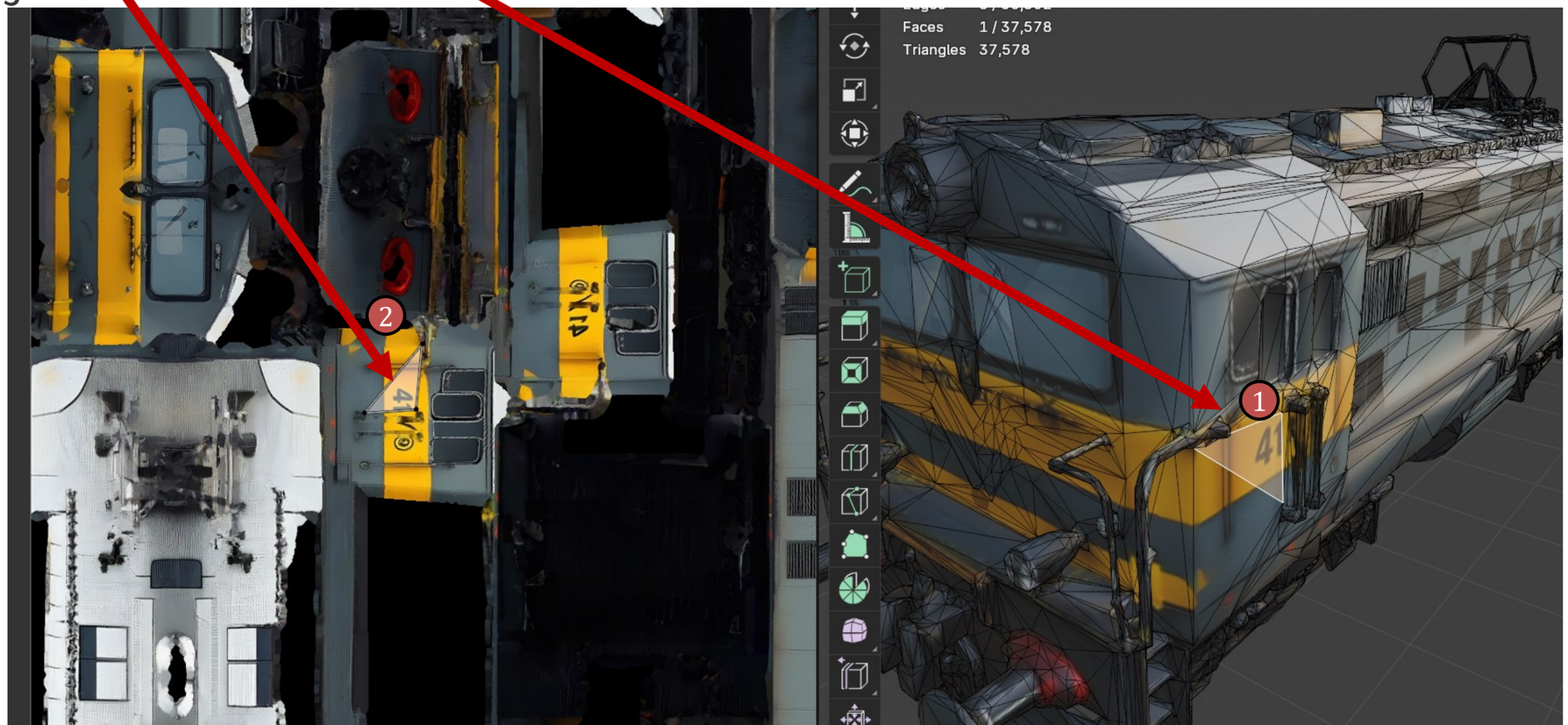
3 



UV Mapping

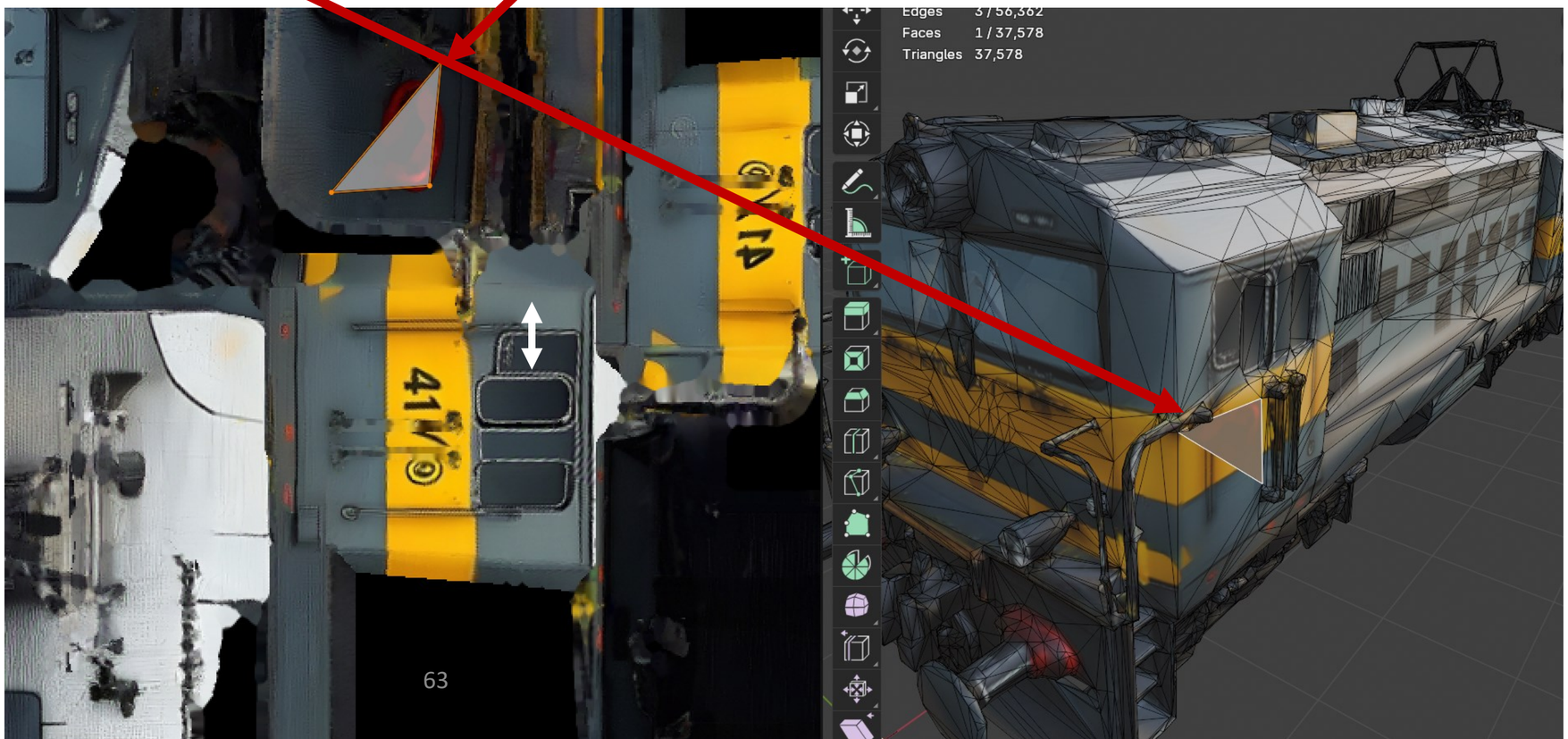
- ❑ Select Face including the Triangle of 41
- ❑ Observe that Triangle on the Image
- ❑ Select several Triangles and See

The links between Mesh and Image



UV Mapping

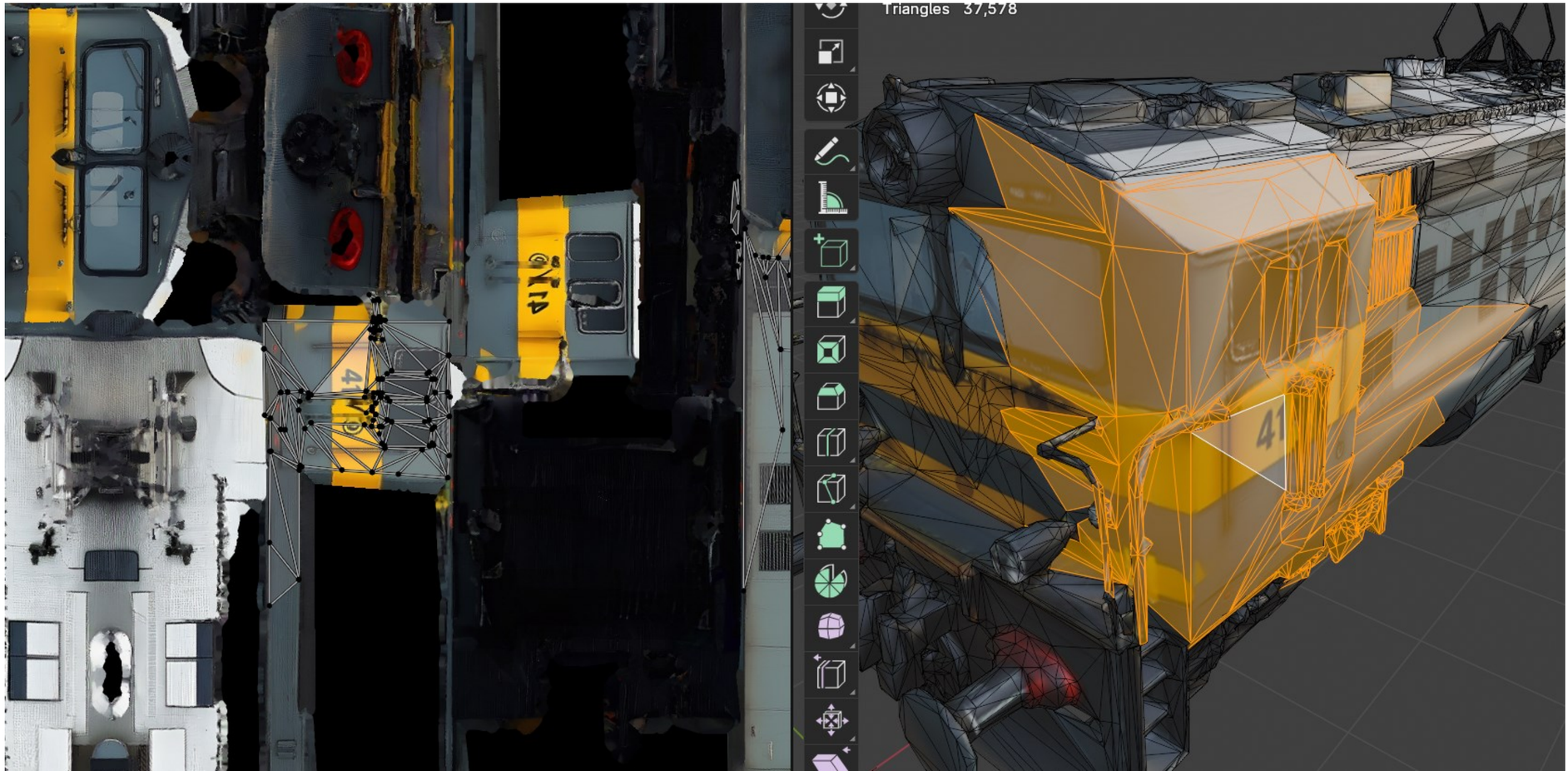
- ❑ In UV Editor → Press “G” and Move the Triangle over the image to the Red light
- ❑ Observe that the Mesh color
- ❑ Undo (Ctrl Z)



UV Mapping

Think & Observe - No submission required

❑ What is UV mapping?



Normal / Face Orientation in Blender

Check whether face normals point outward before export

How to Check

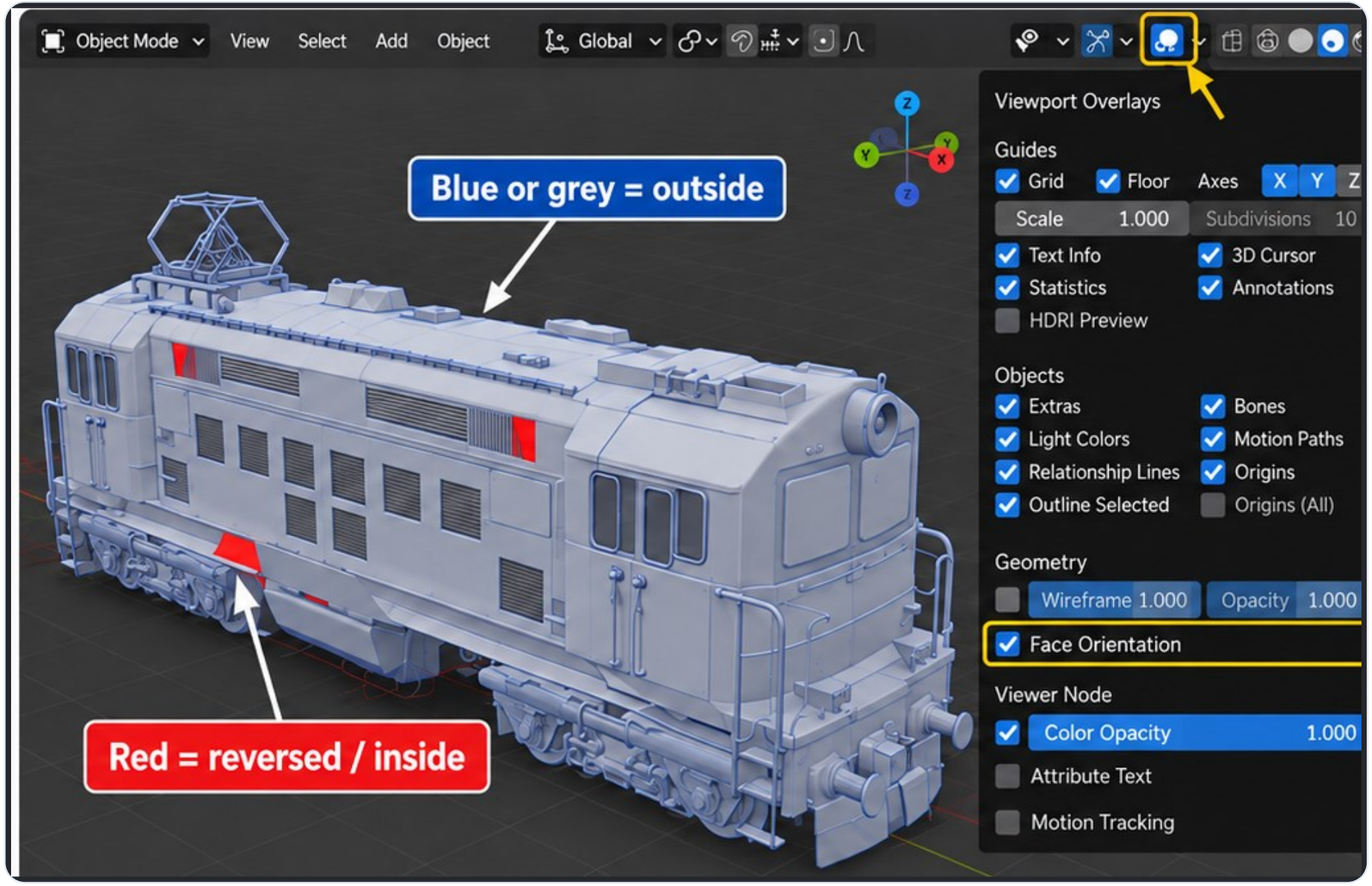
- [ ] Open Viewport Overlays
- [ ] Enable “Face Orientation”
- [ ] Grey or blue = outside surface
- [ ] Red = reversed / inside surface

Why It Matters

- [ ] Red faces may vanish in Unity or Unreal
- [ ] Lighting and shadows may look wrong
- [ ] Materials may display incorrectly
- [ ] The model may look broken after export

How to Fix

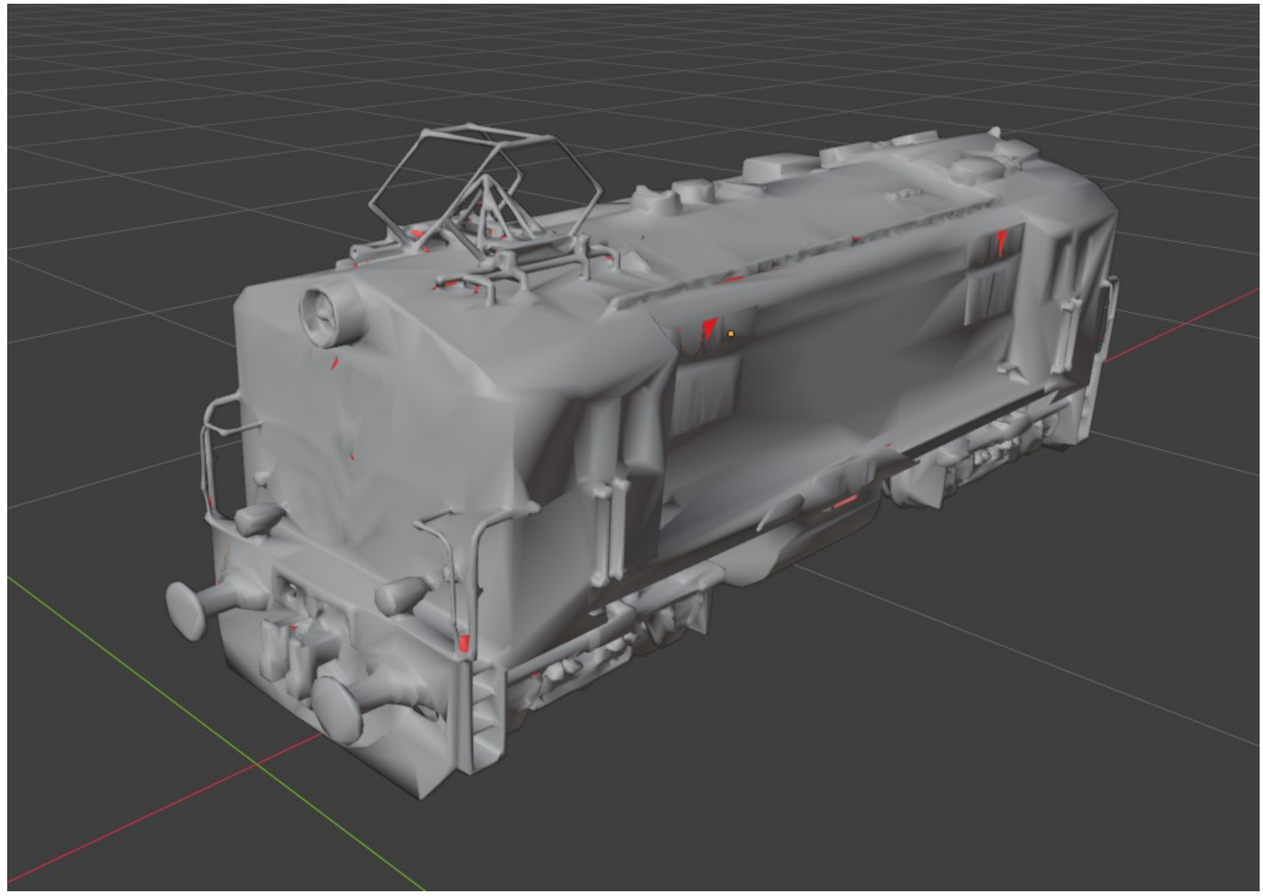
1. Select the object
2. Go to Edit Mode
3. Press A to select all faces
4. Mesh → Normals → Recalculate Outside



Face Normal

We did our best for Face Normal

- ❑ If there are small faces remained still red → you can ignore it



Naming Objects, Meshes, and Materials in Blender

What Are These Three Names?



LocomotiveFreight — **Object** = whole item in the scene

LocomotiveFreight_Mesh — **Mesh** = geometry / shape

LocomotiveFreight_BodyPaint
LocomotiveFreight_Glass — **Material** = appearance / surface

Why Naming Is Important

- Easier to find items in the Outliner
- Cleaner FBX export to Unity / Unreal
- Better troubleshooting and editing
- Easier material assignment and replacement
- Helps teamwork, scripting, and large scenes



Bad vs Good Naming

Bad Names

- Mesh1.0
- Material.004
- Cube
- temp

Good Names

- LocomotiveFreight
- LocomotiveFreight_Mesh
- LocomotiveFreight_BodyPaint
- LocomotiveFreight_Glass

Recommended Naming Rule

Formula: "AssetName", "AssetName_Mesh", "AssetName_MaterialType"

Example:

Object: "LocomotiveFreight"
Mesh: "LocomotiveFreight_Mesh"
Material: "LocomotiveFreight_BodyPaint"
Optional extra material: "LocomotiveFreight_Glass"

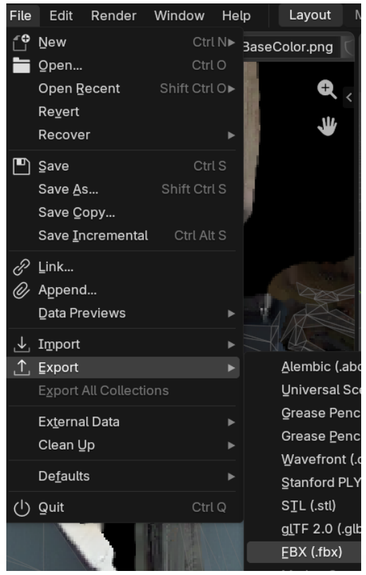
Use clear, descriptive, and consistent names.

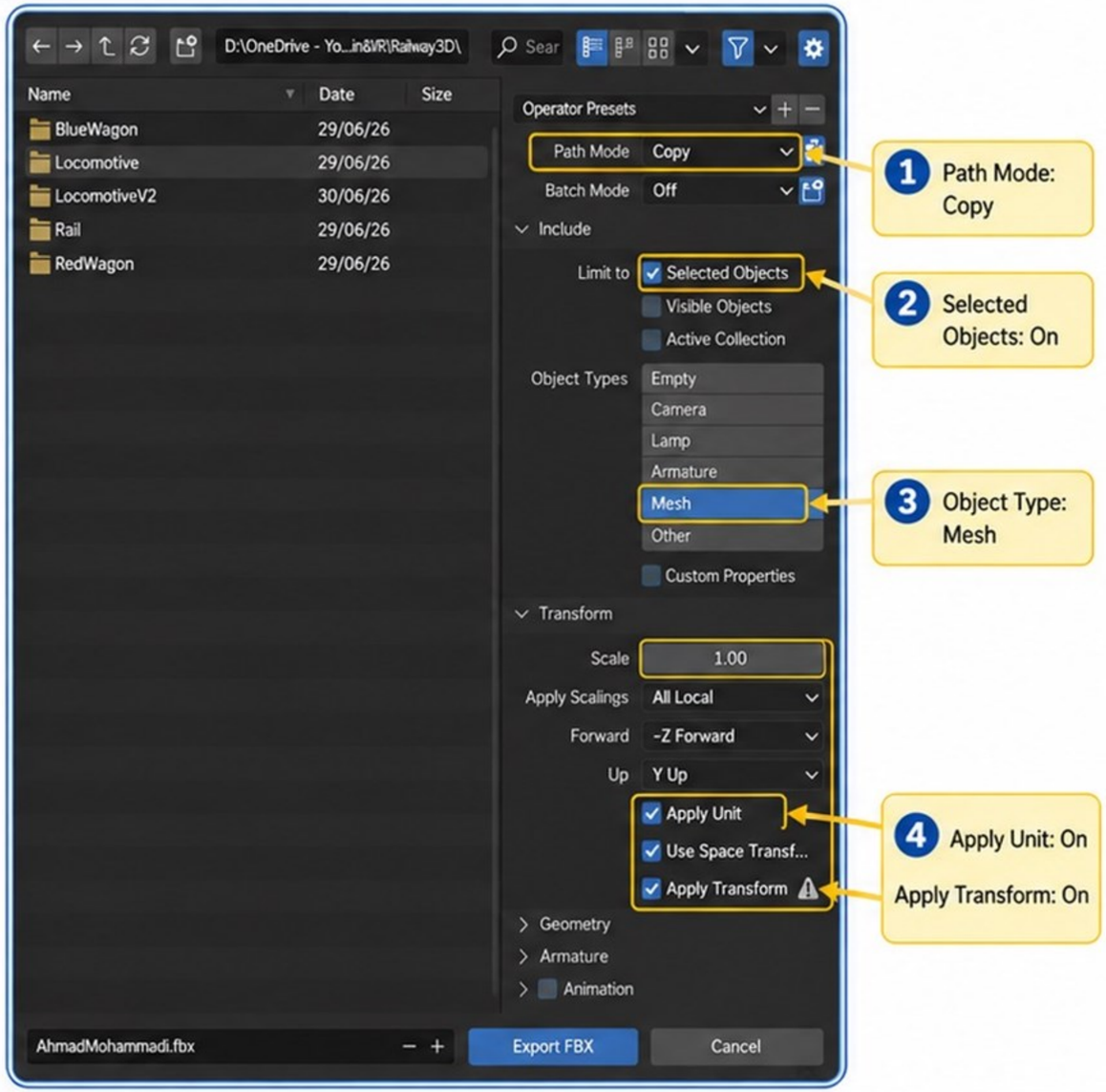
Naming Objects

- Put Proper Names According to Previous Slides

Recommended FBX Export Settings in Blender

- ❑ Select Train model in Viewport
- ❑ File → Export → Export FBX





Why These Settings?

Path Mode: Copy
Copies linked texture/image paths so exported assets are easier to transfer and keep organized.

Selected Objects
Exports only the chosen model, avoiding unwanted objects such as cameras, lights, or helpers.

Object Type: Mesh
Exports only geometry, which is usually what is needed for train, wagon, rail, or building models.

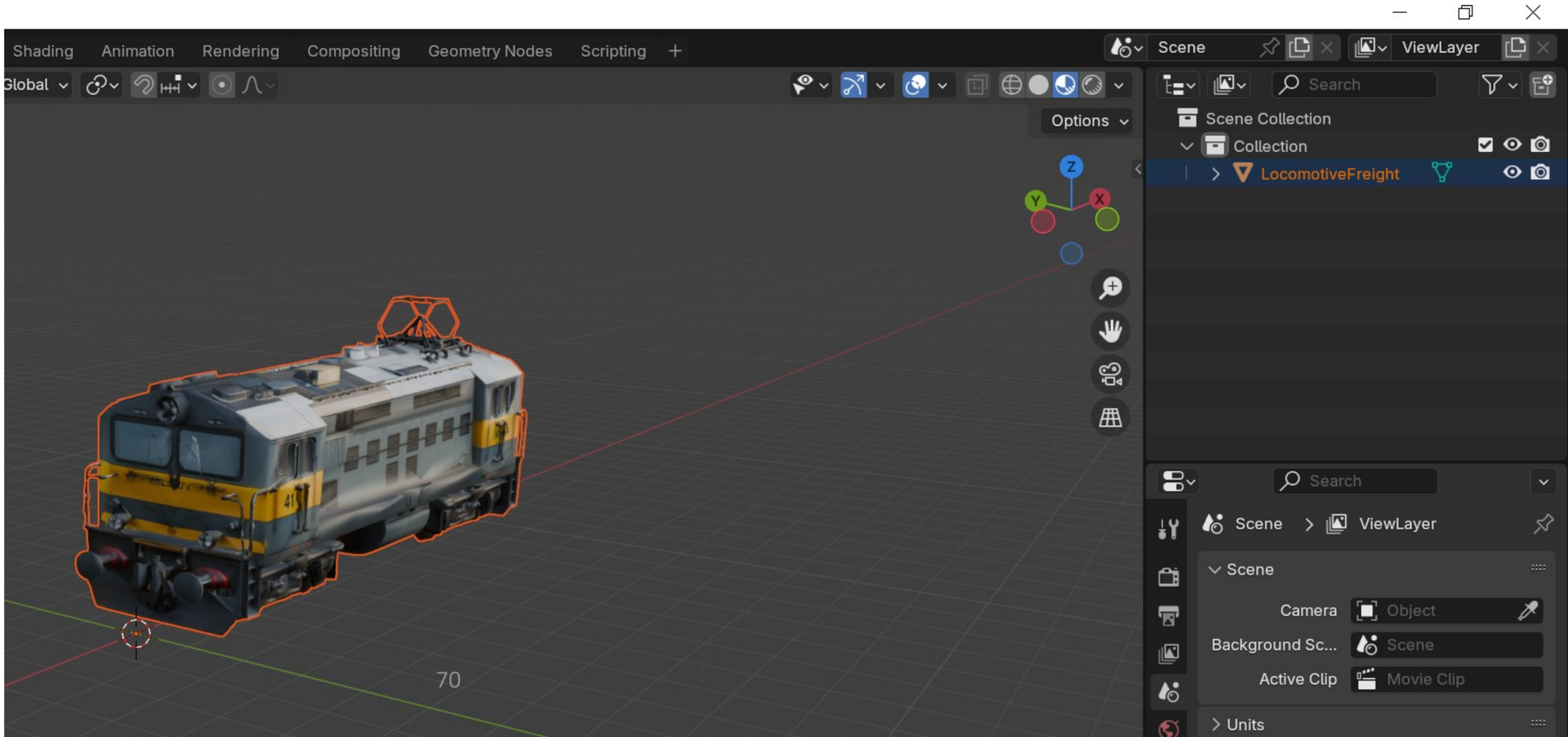
Apply Unit / Apply Transform
Helps preserve correct real-world scale and orientation in Unity/Unreal after export.

Key Benefits

- Cleaner export
- Correct size and orientation
- Fewer import problems
- Easier workflow in game engines

Import The Saved FBX

- ❑ Open a New Blender
- ❑ Import your saved FBX file to check, if it saved properly
- ❑ Check everything is Okay

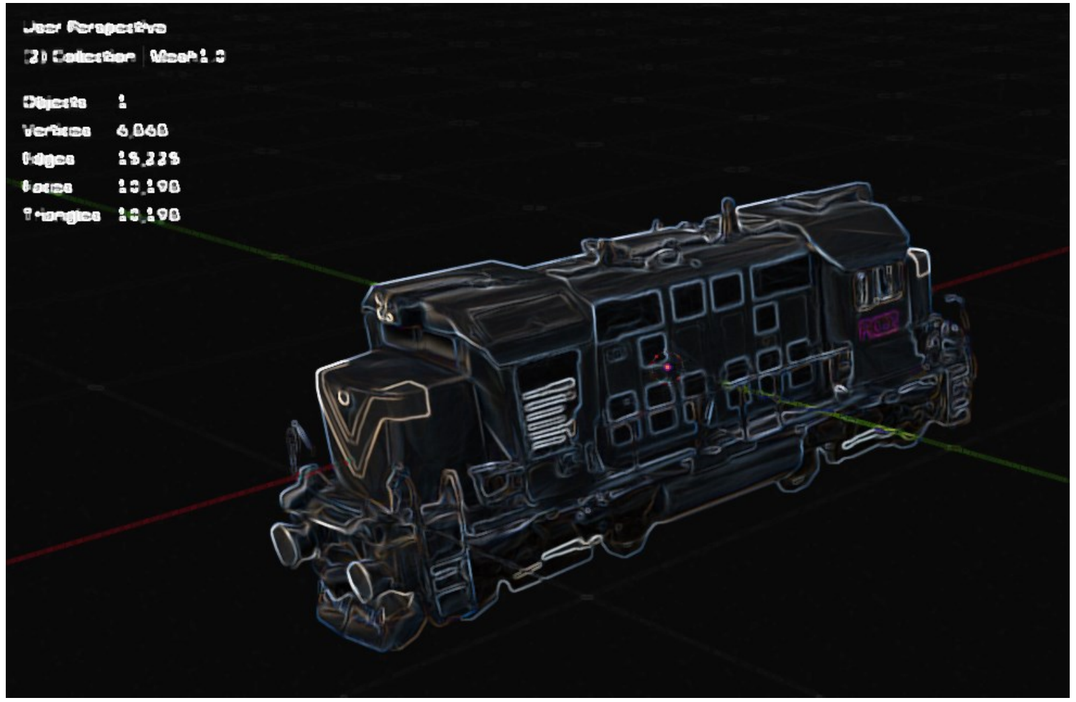


End of Milestone 2

Deliverable 2

- ❑ Open **your exported Locomotive** from Meshy.ai
- ❑ Apply slides of Milestone 2
- ❑ Export Your Model as FBX and Zip it and Submit

Your exported Locomotive from Meshy.ai



Apply the processes

Final Output → Submit This File (fbx) (make it zip)

Beginning of Milestone 3 (Optional)

Combining Meshes

Why Combine Meshes?

- Easier to move, scale, rename, and export
- Cleaner FBX export to Unity / Unreal / digital twin platforms
- Reduces scene complexity
- Makes the object easier to manage

Before: Many Separate Meshes



Outliner (Example)

Body	Light_01
Wheel_FL	Light_02
Wheel_FR	Pipe_01
Wheel_RL	Pipe_02
Wheel_RR	Handrail_01
Window_01	Vent_01
Window_02	Coupler_01
Door_01	...

- ✗ Harder to manage
- ✗ More objects in export
- ✗ Messy hierarchy

Ctrl + J
Join selected meshes

After: One Cleaner Digital Object



Outliner (Example)

- Object: LocomotiveFreight
  - Mesh: LocomotiveFreight_Mesh

- ✓ Cleaner hierarchy
- ✓ Easier export
- ✓ Better for real-time use

Typical Workflow

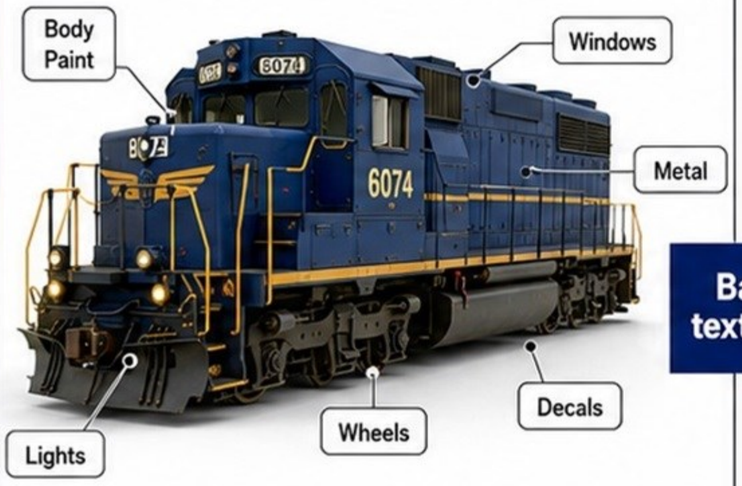
1. Select related mesh parts
2. Keep only the needed geometry
3. Ctrl + J — Press Ctrl + J to join
4. Rename the object clearly
5. Check origin, scale, rotation, and materials

Baking Textures into One Texture

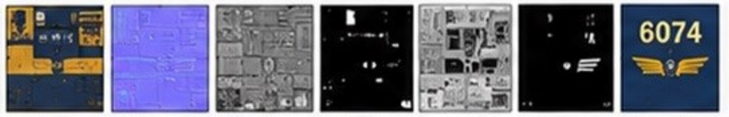
Why Bake Textures?

- Combines many materials into fewer textures
- Reduces draw calls in real-time engines
- Makes export folders cleaner and easier to manage
- Improves performance in VR/XR and digital twins

Before: Many Materials / Textures



Example Texture Maps (Many Files)



Albedo | Normal | Roughness | Metallic | AO | Emissive | Decals

- More material slots
- More texture files
- More draw calls

Bake textures

After: One Baked Texture



Baked Texture (One Atlas)



Material Setup (Simplified)



- Cleaner export
- Fewer materials
- Better real-time performance

Typical Workflow

1. Prepare UV mapping
2. Create a new image texture
3. Bake color/details to one texture
4. Save the baked image
5. Assign the new baked material
6. Export the model with texture files

Level of Detail (LOD)

Why Use LOD?

- Reduces triangle count
- Improves FPS / performance
- Supports larger scenes
- Important for VR/XR and real-time digital twins



Typical Workflow

1. Start with a detailed model
2. Create simpler versions
3. Remove small details for far views
4. Keep overall shape consistent
5. Import into Unity / Unreal
6. Assign to an LOD Group

Learn from a YouTube Tutorial

☐ **Watch the tutorial from the RoadwayVR channel and apply the process step by step.**

This activity strengthens students' online learning, self-guided practice, and step-by-step modeling skills.



https://youtu.be/8khJoTdoOvY

12 Tips in the Tutorial

1. Check licensing of 3D model
2. Organize your file
3. Select proper names for asset and materials/textures
4. Check tris / vertices and optimize if needed
5. Apply all modifiers
6. Combine meshes
7. Apply proper scale
8. Fix coordinate system / location
9. Apply all transforms
10. Set origin to geometry
11. Combine materials / bake textures
12. Create LOD and check face orientation

◔ Average time per tip: about 2 minutes